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Education

Swiss Federal Institute of Technology - Zurich (ETH Zürich) Architecture & Building Systems (A/S) Group Institute of Technology in Architecture (ITA), Department of Architecture (D-ARCH) Doctorate of Sciences (Dr. sc. ETH Zürich) (Ph.D. Equivalent) <i>Dissertation Advisor: Prof. Arno Schlueter, Committee Member: Dr. Stefan Mueller Arisona</i>	Zürich, Switzerland <i>Jan. 2017</i>
National University of Singapore (NUS) Department of the Built Environment Master of Science (MSc. Building) <i>MSc. Thesis Supervisor: Prof. Chandra Sekhar</i>	Singapore <i>Nov. 2011</i>
University of Nebraska - Lincoln (UNL) and Omaha (UNO) Durham School of Architectural Engineering & Construction, Peter Kiewit Institute Master of Architectural Engineering (MAE) Bachelor's of Science in Architectural Engineering (BSAE) <i>MAE Thesis Supervisor: Prof. Gregor Henze</i>	Omaha, NE, USA <i>May 2007</i> <i>May 2006</i>

Academic and Professional Appointments

- **Associate Professor, Singapore Management University (SMU), Singapore**
College of Integrative Studies (CIS)
Sep. 2025 - Present
Tenured Faculty Member and SMU Urban Institute Fellow (Systems), Lee Kong Chian Fellow (2026-2027)
- **Co-Founder and Scientific Advisor, SpaceMatch Pte. Ltd., Singapore**
Aug. 2021 - Present
R&D Advisor and Co-Founder of an NUS Spin-off focused on indoor environmental and spatial optimization
- **Associate Professor, National University of Singapore (NUS), Singapore**
College of Design and Engineering (CDE), Dept. of the Built Environment (DBE)
Jan. 2024 - Aug. 2025
Tenured Faculty Member, Deputy Head of Communication and Engagement (Jul. 2022 - Feb. 2025) and creator of the edX Course, *Data Science for Construction, Architecture and Engineering*, that had 43,500+ participants worldwide enrolled (Apr. 2020 - Dec. 2024)
- **Assistant Professor, National University of Singapore (NUS), Singapore**
Jun. 2017 - Dec. 2023
Tenure-Track Faculty Member
- **Post-Doctoral Researcher, Future Cities Laboratory, Singapore-ETH Centre, Singapore**
Nov. 2016 - Mar. 2017
Performance verification on the 3for2 Project focused on a decentralized ventilation system.
- **Visiting Researcher, Vermont Energy Investment Corporation (VEIC), Burlington, VT, USA**
Jun. - Nov. 2015
Data science guest researcher for analyzing time-series data from large numbers of non-residential buildings.

- **Institute of Technology in Architecture Fellow, ETH Zürich, Zürich, Switzerland**
Oct. 2012 - Oct. 2016
Fully funded 3.5-year competitive Institute of Technology in Architecture Fellowship.
- **Co-Founder and Chief Technology Officer (CTO), Optiras Pte. Ltd., Singapore**
Jul. 2010 - Apr. 2012
Cleantech start-up that was seed-funded by the Singapore NRF and SWG Incubator.
- **Fulbright Scholar and Researcher, National University of Singapore, Singapore**
Jul. 2009 - Jun. 2010
Fulbright research grant to enhance the core EnergyPlus whole building simulation engine.
- **Research and Development Manager, Ezenics, Omaha, NE, USA**
Oct. 2008 - Jun. 2009
Leader of an R&D team focused on connectivity, development, and implementation of plug-n-play solutions

Awards and Honors

Major Global Research Awards and Rankings

- **Best Paper Award at the 4th International Workshop on Social Infrastructure Systems (Social-Sys'26) in Banff, Canada, 2026**, Role: Senior author within a group led by Tom Parkinson, Paper title: *Towards Uncovering Indoor Satisfaction Profiles: LLM Capacity for Structured Extraction from Occupant Feedback*
- **Stanford University World's Top 2% Scientists Ranking based on Career and Annual Performance during 2021-2024, 2022-2025**
- **Young Scholar Award at the Sustainable Built Environment Conference 2025 in Tokyo, Japan, 2025**, Role: Co-Author within a group led by Jean You, Paper title: *Evaluating heat exposure and vulnerability among older adults using wearable technology to support aging in place*
- **Annual Best Paper Award in the Building and Environment Journal for 2022, 2023**, Role: Co-Author within a large group from Annex 79 Subtask 1 led by Giorgia Chinazzo, Paper title: *Quality criteria for multi-domain studies in the indoor environment: Critical review towards research guidelines and recommendations*
- **Best Poster Award, 17th International Conference of the International Society of Indoor Air Quality & Climate (Indoor Air 2022) Conference, Kuopio, Finland 2022**, Role: Lead, and First Author of Conference Paper, Paper Title: *Smartwatch-based ecological momentary assessments for occupant wellness and privacy in buildings*
- **Best Paper Awards in the Buildings Journal for 2020, First Place Award, 2022**, Role: Senior/Last and Corresponding Author, Paper Title: *Humans-as-a-Sensor for buildings — Intensive longitudinal indoor comfort models*

Major Global or University-level Education Awards and Nominations

- **Best Webinar Award at the IBPSA-USA SimBuild 2024 Conference in Denver, USA** for the online webinar entitled *Skills for Tackling Built Environment Challenges* that was given in Jan. 2023
- **NUS Annual Teaching Education Award (ATEA/ADEA) Honour Roll (2024-2028)**
- **NUS Annual Digital Education Awards (ADEA) for 2021, 2022, and 2023**, University-level Award, For AY2019-2020, AY2020-2021, and AY2021-2022 Teaching
- **Finalist for the 2021 Global edX Prize for Exceptional Contributions in Online Teaching and Learning**, International Award, 2021

Fellowships/Scholarships

- **ETH Zürich Architecture and Technology Fellowship, 2012-2016**
- **ASHRAE/IBPSA Conference Student Travel Grant, 2014**
- **National University of Singapore (NUS) Research Scholarship, 2010**
- **U.S. Fulbright Student Fellowship to the Republic of Singapore, 2009-2010**
- **ASHRAE Nebraska Chapter Scholarship, 2005-2006**
- **Walter Scott Jr. Scholarship, 2002-2006**

Research

Peer-reviewed Publications in International Journals

* = Corresponding Author

1. Alva P*, Talami R, Pei W, Sibenik G, Mosteiro-Romero M, **Miller C**, Stouffs R. *Potential recyclable materials in buildings: A framework for greenhouse gas emissions assessment of residential buildings in Singapore*. Resour Conserv Recycl. 2026;232(108954):108954. <http://dx.doi.org/10.1016/j.resconrec.2026.108954>
2. Bayer DR*, Hu M, **Miller C**, Pruckner M. *Optimizing heat pump and electric vehicle flexibility through predictive home energy management systems within urban energy digital twins*. Energy Build. 2026;367(117674):117674. <http://dx.doi.org/10.1016/j.enbuild.2026.117674>
3. Parkinson T, Bhangar S, Chan WR, Chen W, Gilligan B, Kramer T, Li J, **Miller C**, Oaks L, Thind MP, Licina D*. *Ten questions concerning low-cost indoor air quality sensors: Perspectives from research and practice*. Build Environ. 2026;300(114737):114737. <http://dx.doi.org/10.1016/j.buildenv.2026.114737>
4. Gu H, Jin X, Xiao F*, Hu M, **Miller C**. *An ex-post analysis framework for assessing the effectiveness and driving factors of green building certification adoption: A case study of Singapore*. Energy Build. 2026;351(116733):116733. <http://dx.doi.org/10.1016/j.enbuild.2025.116733>
5. Cheng S, Lei B, Fujiwara K, **Miller C**, Biljecki F*, van Ameijde J. *Walking through green and grey: Exploring sequential exposure and multisensory environmental effects on psychological restoration*. Build Environ. 2026;287(113748):113748. <http://dx.doi.org/10.1016/j.buildenv.2025.113748>
6. Lei B, Liu P, Fujiwara K, Frei M, **Miller C**, Chua YX, Biljecki F*. *Multidimensional analysis of human outdoor comfort: Integrating just-in-time adaptive interventions (JITAI) in urban digital twins*. Cities. 2026;168(106443):106443. <http://dx.doi.org/10.1016/j.cities.2025.106443>
7. Lien SK*, Canaydin A, **Miller C**, Fu C, Kazmi H, Rajasekharan J. *Cross-domain disaggregation of electricity for heating in all-electric school buildings – learning from school buildings with district heating*. Energy Build. 2025;348(116359):116359. <http://dx.doi.org/10.1016/j.enbuild.2025.116359>
8. Vittori F, Tan CF, Pisello AL*, Chong A, Piselli C, **Miller C**. *BIM-to-BRICK: Using graph modeling for IoT/BMS and spatial semantic data interoperability within digital data models of buildings*. Energy Build. 2025;348(116368):116368. <http://dx.doi.org/10.1016/j.enbuild.2025.116368>
9. Rehmann F*, Mosteiro-Romero M, **Miller C**, Streblow R. *Enhancing urban energy modeling: A case study of data acquisition, enrichment, and evaluation in Berlin*. Energy Build. 2025;116070. <http://dx.doi.org/10.1016/j.enbuild.2025.116070>
10. **Miller C***, Chua YX, Quintana M, Lei B, Biljecki F, Frei M. *Make yourself comfortable: Nudging urban heat and noise mitigation with smartwatch-based Just-in-time Adaptive Interventions (JITAI)*. Build Environ. 2025;284(113388):113388. <http://dx.doi.org/10.1016/j.buildenv.2025.113388>
11. Yap W, Wu AN, **Miller C**, Biljecki F*. *Revealing building operating carbon dynamics for multiple cities*. Nat Sustain. 2025. <http://dx.doi.org/10.1038/s41893-025-01615-8>
12. Zhong S*, Chwalek P, Perry N, Ramsay D, **Miller C**, Lalanne D, Alavi HS, Paradiso JA. *Sensors and sensibilities: Exploring interactions for habitat comfort with an environmental-physiological sensing eyewear in the wild*. Int J Hum Comput Stud. 2025;200(103510):103510. <http://dx.doi.org/10.1016/j.ijhcs.2025.103510>
13. Abdelrahman M, Macatulad E, Lei B, Quintana M, **Miller C**, Biljecki F*. *What is a Digital Twin anyway? Deriving the definition for the built environment from over 15,000 scientific publications*. Build Environ. 2025;274(112748):112748. <http://dx.doi.org/10.1016/j.buildenv.2025.112748>
14. Zhang W, Quintana M, **Miller C***. *Recommender systems and reinforcement learning for human-building interaction and context aware support: A text mining-driven review of scientific literature*. Energy Build. 2025;329(115247):115247. <http://dx.doi.org/10.1016/j.enbuild.2024.115247>
15. Daemei AB*, Lovreglio R, Feng Z, Paes D, **Miller C**. *Gamification for air quality education: A systematic literature review*. Build Environ. 2025;270(112526):112526. <http://dx.doi.org/10.1016/j.buildenv.2025.112526>
16. Chwalek P*, Zhong S, Perry N, Liu T, **Miller C**, Alavi HS, Lalanne D, Paradiso J. *A dataset exploring urban comfort through novel wearables and environmental surveys*. Sci Data. 2024;11(1):1423. <http://dx.doi.org/10.1038/s41597-024-04279-9>
17. Dabirian S*, **Miller C**, Adli A, Eicker U. *Gaussian-based plug load profile prediction in non-residential buildings archetype*. Appl Energy. 2024;374: 123970. <http://dx.doi.org/10.1016/j.apenergy.2024.123970>

18. Alva P*, Mosteiro-Romero M, **Miller C**, Stouffs R. *Mitigating operational greenhouse gas emissions in ageing residential buildings using an Urban Digital Twin dashboard*. Energy Build. 2024;322(114681):114681. <http://dx.doi.org/10.1016/j.enbuild.2024.114681>
19. Fu C, Kazmi H, Quintana M, **Miller C***. *Creating synthetic energy meter data using conditional diffusion and building metadata*. Energy Build. 2024;312: 114216. <http://dx.doi.org/10.1016/j.enbuild.2024.114216>
20. Mosteiro-Romero M*, Quintana M, Stouffs R, **Miller C**. *A data-driven agent-based model of occupants' thermal comfort behaviors for the planning of district-scale flexible work arrangements*. Build Environ. 2024;257: 111479. <http://dx.doi.org/10.1016/j.buildenv.2024.111479>
21. Liguori A*, Quintana M, Fu C, **Miller C**, Frisch J, van Treeck C. *Opening the Black Box: Towards inherently interpretable energy data imputation models using building physics insight*. Energy Build. 2024;310: 114071. <http://dx.doi.org/10.1016/j.enbuild.2024.114071>
22. Canaydin A, Fu C, Balint A, Khalil M, **Miller C**, Kazmi H*. *Interpretable domain-informed and domain-agnostic features for supervised and unsupervised learning on building energy demand data*. Appl Energy. 2024;360: 122741. <http://dx.doi.org/10.1016/j.apenergy.2024.122741>
23. Martin M, Ramani V, **Miller C***. *InfraRed Investigation in Singapore (IRIS) Observatory: Urban heat island contributors and mitigators analysis using neighborhood-scale thermal imaging*. Energy Build. 2024;307: 113973. <http://dx.doi.org/10.1016/j.enbuild.2024.113973>
24. Ramani V, Arjunan P, Poolla K, **Miller C***. *Semantic segmentation of longitudinal thermal images for identification of hot and cool spots in urban areas*. Build Environ. 2024;249: 111112. <http://dx.doi.org/10.1016/j.buildenv.2023.111112>
25. Lei Y, Tekler ZD, Zhan S, **Miller C**, Chong A*. *Experimental evaluation of thermal adaptation and transient thermal comfort in a tropical mixed-mode ventilation context*. Build Environ. 2024;248:111043. <http://dx.doi.org/10.1016/j.buildenv.2023.111043>
26. Hepf C*, Gottkehas Kamp B, **Miller C**, Auer T. *International comparison of weather and emission predictive building control*. Buildings. 2024;14: 288. <http://dx.doi.org/10.3390/buildings14010288>
27. Fu C, Quintana M, Nagy Z, **Miller C***. *Filling time-series gaps using image techniques: Multidimensional context autoencoder approach for building energy data imputation*. Appl Therm Eng. 2024;236: 121545. <http://dx.doi.org/10.1016/j.applthermaleng.2023.121545>
28. Lin S, Ramani V, Martin M, Arjunan P, Chong A, Biljecki F, Ignatius M, Poolla K, **Miller C***. *District-scale surface temperatures generated from high-resolution longitudinal thermal infrared images*. Sci Data. 2023;10: 859. <http://dx.doi.org/10.1038/s41597-023-02749-0>
29. Lorenz C-L*, André M, Abele O, Gunay B, Hahn J, Hensen P, Nagy Z, Ouf MM, Park JY, Yaduvanshi NS, **Miller C**. *A repository of occupant-centric control case studies: Survey development and database overview*. Energy Build. 2023;300: 113649. <http://dx.doi.org/10.1016/j.enbuild.2023.113649>
30. Nagy Z*, Gunay B, **Miller C**, Hahn J, Ouf M, Lee S, Hobson B, Abuimara T, Bandurski K, André M, Lorenz C-L, Crosby S, Dong B, Jiang Z, Peng Y, Favero M, Park J-Y, Nweye K, Nojedehi P, Stopps H, Sarran L, Brackley C, Bassett K, Govertsen K, Koczorek N, Abele O, Casavant E, Kane M, O'Neill Z, Yang T, Day J, Huchuk B, Hellwig R, Vellei M. *Ten questions concerning occupant-centric control and operations*. Build Environ. 2023; 110518. <http://dx.doi.org/10.1016/j.buildenv.2023.110518>
31. Kazmi H*, Fu C, **Miller C**. *Ten questions concerning data-driven modelling and forecasting of operational energy demand at building and urban scale*. Build Environ. 2023;239: 110407. <http://dx.doi.org/10.1016/j.buildenv.2023.110407>
32. Peng Y, Antanuri N, Lau S-K, Jebelli B, Jusuf SK, **Miller C**, Teo YT, Chua YX, Chong A*. *Experimental assessment of thermal and acoustics interactions on occupant comfort in mixed-mode buildings*. Build Environ. 2023;238: 110342. <http://dx.doi.org/10.1016/j.buildenv.2023.110342>
33. Liu P, Zhao T, Luo J, Lei B, Frei M, **Miller C**, Biljecki F*. *Towards Human-centric Digital Twins: Leveraging Computer Vision and Graph Models to Predict Outdoor Comfort*. Sustainable Cities and Society. 2023;93: 104480. <http://dx.doi.org/10.1016/j.scs.2023.104480>
34. Mosteiro-Romero M*, **Miller C**, Chong A, Stouffs R. *Elastic buildings: Calibrated district-scale simulation of occupant-flexible campus operation for hybrid work optimization*. Build Environ. 2023;237: 110318. <http://dx.doi.org/10.1016/j.buildenv.2023.110318>
35. Ramani V, Martin M, Arjunan P, Chong A, Poolla K, **Miller C***. *Longitudinal thermal imaging for scalable non-residential HVAC and occupant behaviour characterization*. Energy Build. 2023;287: 112997. <http://dx.doi.org/10.1016/j.enbuild.2023.112997>

36. Jin X, Zhang C, Xiao F*, Li A, **Miller C**. *A review and reflection on open datasets of city-level building energy use and their applications*. Energy Build. 2023;285: 112911. <http://dx.doi.org/10.1016/j.enbuild.2023.112911>
37. Tekler ZD, Lei Y, Peng Y, **Miller C**, Chong A*. *A hybrid active learning framework for personal thermal comfort models*. Build Environ. 2023;234: 110148. <http://dx.doi.org/10.1016/j.buildenv.2023.110148>
38. Teitelbaum E*, **Miller C**, Meggers F. *Highway to the Comfort Zone: History of the Psychrometric Chart*. Buildings. 2023;13: 797. <http://dx.doi.org/10.3390/buildings13030797>
39. Quintana M, Schiavon S, Tartarini F, Kim J, **Miller C***. *Cohort comfort models — Using occupant’s similarity to predict personal thermal preference with less data*. Build Environ. 2023;227: 109685. <http://dx.doi.org/10.1016/j.buildenv.2022.109685>
40. Tartarini F*, Schiavon S, Quintana M, **Miller C**. *Personal comfort models based on a 6-month experiment using environmental parameters and data from wearables*. Indoor Air. 2022;32: e13160. <http://dx.doi.org/10.1111/ina.13160>
41. Chinazzo G*, Andersen RK, Azar E, Barthelmes VM, Becchio C, Belussi L, Berger C, Carlucci S, Corgnati SP, Crosby S, Danza L, de Castro L, Favero M, Gauthier S, Hellwig RT, Jin Q, Kim J, Khanie MS, Khovalyg D, Lingua C, Luna-Navarro A, Mahdavi A, **Miller C**, Mino-Rodriguez I, Pigliautile I, Pisello AL, Rupp RF, Sadick AM, Salamone F, Schweiker M, Syndicus M, Spigliantini G, Vasquez NG, Vakalis D, Vellei M, Wei S. *Quality criteria for multi-domain studies in the indoor environment: Critical review towards research guidelines and recommendations*. Build Environ. 2022;226: 109719. <http://dx.doi.org/10.1016/j.buildenv.2022.109719>
42. Pantelic J*, Nazarian N, **Miller C**, Meggers F, Lee JKW, Licina D. *Transformational IoT sensing for air pollution and thermal exposures*. Frontiers in Built Environment. 2022;8. <http://dx.doi.org/10.3389/fbuil.2022.971523>
43. Martin M, Chong A, Biljecki F, **Miller C***. *Infrared thermography in the built environment: A multi-scale review*. Renewable Sustainable Energy Rev. 2022; 165: 112540. <http://dx.doi.org/10.1016/j.rser.2022.112540>
44. Quintana M, Stoeckmann T, Park JY, Turowski M, Hagenmeyer V, **Miller C***. *ALDI++: Automatic and parameter-less discord and outlier detection for building energy load profiles*. Energy Build. 2022; 265: 112096. <http://dx.doi.org/10.1016/j.enbuild.2022.112096>
45. Salamone F*, Chinazzo G, Danza L, **Miller C**, Sibilio S, Masullo M. *Low-cost thermohygrometers to assess thermal comfort in the built environment: A laboratory evaluation of their measurement performance*. Buildings. 2022; 12: 579. <http://dx.doi.org/10.3390/buildings12050579>
46. Leprince J*, Madsen H, **Miller C**, Real JP, van der Vlist R, Basu K, Zeiler W. *Fifty shades of grey: Automated stochastic model identification of building heat dynamics*. Energy Build. 2022; 266: 112095. <http://dx.doi.org/10.1016/j.enbuild.2022.112095>
47. Abdelrahman MM, **Miller C***. *Targeting occupant feedback using digital twins: Adaptive spatial-temporal thermal preference sampling to optimize personal comfort models*. Build Environ. 2022; 218: 109090. <http://dx.doi.org/10.1016/j.buildenv.2022.109090>
48. Kazmi H*, Keijsers M, Mehmood F, **Miller C**. *Energy balances, thermal performance, and heat stress: Disentangling occupant behaviour and weather influences in a Dutch net-zero energy neighborhood*. Energy Build. 2022; 263: 112020. <http://dx.doi.org/10.1016/j.enbuild.2022.112020>
49. **Miller C***, Picchetti B, Fu C, Pantelic J. *Limitations of machine learning for building energy prediction: ASHRAE Great Energy Predictor III Kaggle competition error analysis*. Science and Technology for the Built Environment. 2022; 1–18. <http://dx.doi.org/10.1080/23744731.2022.2067466>
50. Fu C, **Miller C***. *Using Google Trends as a proxy for occupant behavior to predict building energy consumption*. Appl Energy. 2022; 310: 118343. <http://dx.doi.org/10.1016/j.apenergy.2021.118343>
51. Abdelrahman MM, Chong A, **Miller C***. *Personal thermal comfort models using digital twins: Preference prediction with BIM-extracted spatial-temporal proximity data from Build2Vec*. Build Environ. 2022; 207: 108532. <http://dx.doi.org/10.1016/j.buildenv.2021.108532>
52. Arjunan P*, Poolla K, **Miller C**. *BEEM: Data-driven building energy bEnchMarking for Singapore*. Energy Build. 2022; 111869. <http://dx.doi.org/10.1016/j.enbuild.2022.111869>
53. Leprince J*, **Miller C**, Zeiler W. *Data mining cubes for buildings, a generic framework for multidimensional analytics of building performance data*. Energy Build. 2021; 248: 111195. <http://dx.doi.org/10.1016/j.enbuild.2021.111195>

54. Gunay B*, Nagy Z, **Miller C**, Ouf M, Dong B. *Using Occupant-Centric Control for Commercial HVAC Systems*. ASHRAE Journal. 2021;63: 30–32. <http://bit.ly/3X1TZyb>
55. Roth J, Chadalawada J, Jain RK, **Miller C***. *Uncertainty Matters: Bayesian probabilistic forecasting for residential smart meter prediction, segmentation, and behavioral measurement and verification*. Energies. 2021; 14: 1481. <http://dx.doi.org/10.3390/en14051481>
56. Abdelrahman MM, Zhan S, **Miller C**, Chong A*. *Data science for building energy efficiency: A comprehensive text-mining driven review of scientific literature*. Energy Build. 2021; 110885. <http://dx.doi.org/10.1016/j.enbuild.2021.110885>
57. Joshi A*, **Miller C**. *Review of machine learning techniques for mosquito control in urban environments*. Ecol Inform. 2021; 61: 101241. <http://dx.doi.org/10.1016/j.ecoinf.2021.101241>
58. Periyakoil D*, Das HP, **Miller C**, Spanos CJ, Prata N. *Environmental exposures in Singapore schools: An ecological study*. Int J Environ Res Public Health. 2021; 18: 1843. <http://dx.doi.org/10.3390/ijerph18041843>
59. Quintana M, Arjunan P, **Miller C***. *Islands of misfit buildings: Detecting uncharacteristic electricity use behavior using load shape clustering*. Build Simul. 2021; 14: 119–130. <http://dx.doi.org/10.1007/s12273-020-0626-1>
60. Nazarian N*, Liu S, Kohler M, Lee JKW, **Miller C**, Chow WTL, Alhadad SB, Martelli A, Quintana M, Sundén L, Norford L. *Project Coolbit: Can your watch predict heat stress and thermal comfort sensation?* Environ Res Lett. 2021; 16: 034031. <http://dx.doi.org/10.1088/1748-9326/abd130>
61. Jayathissa P, Quintana M, Abdelrahman M, **Miller C***. *Humans-as-a-Sensor for buildings — Intensive longitudinal indoor comfort models*. Buildings. 2020; 10: 174. <http://dx.doi.org/10.3390/buildings10100174>
62. Roth J*, Martin A, **Miller C**, Jain RK. *SynCity: Using open data to create a synthetic city of hourly building energy estimates by integrating data-driven and physics-based methods*. Appl Energy. 2020; 280: 115981. <http://dx.doi.org/10.1016/j.apenergy.2020.115981>
63. **Miller C***, Arjunan P, Kathirgamanathan A, Fu C, Roth J, Park JY, Balbach C, Gowri K, Nagy Z, Fontanini A, Haberl J. *The ASHRAE Great Energy Predictor III competition: Overview and results*. Science and Technology for the Built Environment. 2020; 1–21. <http://dx.doi.org/10.1080/23744731.2020.1795514>
64. **Miller C***, Kathirgamanathan A, Picchetti B, Arjunan P, Park JY, Nagy Z, Raftery P, Hobson BW, Shi Z, Meggers F. *The Building Data Genome Project 2, energy meter data from the ASHRAE Great Energy Predictor III competition*. Scientific Data. 2020; 7: 368. <http://dx.doi.org/10.1038/s41597-020-00712-x>
65. Arjunan P, Poolla K, **Miller C***. *EnergyStar++: Towards more accurate and explanatory building energy benchmarking*. Appl Energy. 2020; 276: 115413. <http://dx.doi.org/10.1016/j.apenergy.2020.115413>
66. O'Brien W*, Wagner A, Schweiker M, Mahdavi A, Day J, Kjærgaard MB, Carlucci S, Dong B, Tahmasebi F, Yan D, Hong T, Gunay HB, Nagy Z, **Miller C**, Berger C. *Introducing IEA EBC annex 79: Key challenges and opportunities in the field of occupant-centric building design and operation*. Build Environ. 2020; 178: 106738. <http://dx.doi.org/10.1016/j.buildenv.2020.106738>
67. Wang CK*, Tindemans S, **Miller C**, Agugiaro G, Stoter J. *Bayesian calibration at the urban scale: a case study on a large residential heating demand application in Amsterdam*. Journal of Building Performance Simulation. 2020; 13: 347–361. <http://dx.doi.org/10.1080/19401493.2020.1729862>
68. Teitelbaum E*, Jayathissa P, **Miller C**, Meggers F. *Design with Comfort: Expanding the psychrometric chart with radiation and convection dimensions*. Energy and Buildings. 2020. p. 109591. <http://dx.doi.org/10.1016/j.enbuild.2019.109591>
69. Sood T, Janssen P, **Miller C***. *Spacematch: Using environmental preferences to match occupants to suitable activity-based workspaces*. Frontiers in Built Environment. 2020; 6: 113. <http://dx.doi.org/10.3389/fbuil.2020.00113>
70. **Miller C***. *What's in the box?! Towards explainable machine learning applied to non-residential building smart meter classification*. Energy Build. 2019; 199: 523–536. <http://dx.doi.org/10.1016/j.enbuild.2019.07.019>
71. **Miller C***. *More Buildings Make More Generalizable Models—Benchmarking Prediction Methods on Open Electrical Meter Data*. Machine Learning and Knowledge Extraction. 2019; 1: 974–993. <http://dx.doi.org/10.3390/make1030056>
72. Park JY, Yang X, **Miller C**, Arjunan P, Nagy Z*. *Apples or oranges? Identification of fundamental load shape profiles for benchmarking buildings using a large and diverse dataset*. Appl Energy. 2019; 236: 1280–1295. <http://dx.doi.org/10.1016/j.apenergy.2018.12.025>

73. **Miller C***, Meggers F. *Mining electrical meter data to predict principal building use, performance class, and operations strategy for hundreds of non-residential buildings*. Energy Build. 2017; 156: 360–373. <http://dx.doi.org/10.1016/j.enbuild.2017.09.056>
74. **Miller C***, Nagy Z, Schlueter A. *A review of unsupervised statistical learning and visual analytics techniques applied to performance analysis of non-residential buildings*. Renewable Sustainable Energy Rev. 2018; 81: 1365–1377. <http://dx.doi.org/10.1016/j.rser.2017.05.124>
75. **Miller C***, Thomas D, Kampf J, Schlueter A. *Urban and building multiscale co-simulation: case study implementations on two university campuses*. Journal of Building Performance Simulation. 2018; 11: 309–321. <http://dx.doi.org/10.1080/19401493.2017.1354070>
76. Pantelic J, Rysanek A, **Miller C**, Peng Y, Teitelbaum E, Meggers F, Schlueter A*. *Comparing the indoor environmental quality of a displacement ventilation and passive chilled beam application to conventional air-conditioning in the Tropics*. Build Environ. 2018; 130: 128–142. <http://dx.doi.org/10.1016/j.buildenv.2017.11.026>
77. Schlueter A*, Rysanek A, Meggers F, Mast, M, Bruelisauer M, Chen KW, **Miller C**, Pantelic J. *3for2: realizing spatial, material, and energy savings through integrated design*. CTBUH Journal. 2016; 40–45. <https://www.jstor.org/stable/90006203>
78. **Miller C***, Nagy Z, Schlueter A. *Automated daily pattern filtering of measured building performance data*. Autom Constr. 2015;4 9: 1–17. <http://dx.doi.org/10.1016/j.autcon.2014.09.004>
79. Nagy Z*, Rossi D, Hersberger C, Irigoyen SD, **Miller C**, Schlueter A. *Balancing envelope and heating system parameters for zero emissions retrofit using building sensor data*. Appl Energy. 2014; 131: 56–66. <http://dx.doi.org/10.1016/j.apenergy.2014.06.024>
80. Bruelisauer M, Chen KW, Iyengar R, Leibundgut H, Li C, Li M, Mast M*, Meggers F, **Miller C**, Rossi D, Saber EM, Tham KW, Schlueter A. *BubbleZERO—Design, Construction and Operation of a Transportable Research Laboratory for Low Exergy Building System Evaluation in the Tropics*. Energies. 2013; 6: 4551–4571. <http://dx.doi.org/10.3390/en6094551>

Peer-reviewed Publications in International Conferences

1. Zhang W*, Chong A, Schiavon S, **Miller C**. *Scalable adaptive AC control in real sleep environments: A pilot deployment with lightweight reinforcement learning*. Proceedings of the 2026 ACM Sustainability Week. ACM; 2026:370-374. <https://doi.org/10.1145/3765611.3813761>
2. Fu C*, Cheng LW, Chuang TH, Kazmi H, **Miller C**. *Zero-shot building energy forecasting using Chronos-2 foundation models*. Proceedings of the 2026 ACM Sustainability Week. ACM; 2026:448-452. <https://doi.org/10.1145/3765611.3815488>
3. Parkinson T*, Zhang W, Schiavon S, **Miller C**. *Towards uncovering indoor satisfaction profiles: LLM capacity for structured extraction from occupant feedback*. Proceedings of the 2026 ACM Sustainability Week. ACM; 2026:525-529. <https://doi.org/10.1145/3765611.3815502>
4. You J*, Chan JH, Stouffs R, Gottkehas Kamp BG, **Miller C**. *Evaluating heat exposure and vulnerability among older adults using wearable technology to support aging in place*. IOP Conf Ser Earth Environ Sci. 2026;1582(1):012052. <https://doi.org/10.1088/1755-1315/1582/1/012052>
5. Kashiwazaki M*, Takemoto S, Fukawa Y, Shimoyama C, Minami K, Watanabe F, Frei M, Tartarini F, **Miller C**, Schiavon S, Tanabe SI. *Longitudinal evaluation of office work performance and well-being with a focus on individual differences*. Lecture Notes in Civil Engineering. https://doi.org/10.1007/978-3-032-06810-1_1
6. Tolnai BA*, Zimmermann RS, Xie Y, Tran, N, Çeliker, CE, Ma Z, Sartori, I, Amos M, Bengtsson, G, Lien SK, **Miller C**, Hameed A, Jørgensen BN. *Advancing non-intrusive load monitoring: Insights from the winning algorithms in the ADRENALIN 2024 load disaggregation competition*. Lecture Notes in Computer Science. Vol 16096. 2026:338-354. https://doi.org/10.1007/978-3-032-03098-6_22
7. Tolnai BA*, Ma Z, Sartori I, **Miller C**, White S, Amos M, Bengtsson G, Hameed A, Jørgensen BN. *Lessons learned from the ADRENALIN load disaggregation challenge*. Lecture Notes in Computer Science. Vol 16096. 2026:371-387. https://doi.org/10.1007/978-3-032-03098-6_24
8. Zhang W*, Schiavon S, **Miller C**. *Poster Abstract: Field evaluation of body thermoregulation-based dynamic bedroom air temperature control to improve sleep*. BuildSys '25: Proceedings of the 12th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation. ACM; 2025:294-295. <https://doi.org/10.1145/3736425.3772103>

9. Demir N*, Qaiser T, Canaydin A, Fu C, Quintana M, **Miller C**, Kazmi H. *How realistic are synthetic energy data? Feature-based evaluation of generated smart meter load profiles*. BuildSys '25: Proceedings of the 12th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation. ACM; 2025:461-464. <https://doi.org/10.1145/3736425.3772358>
10. Alva P*, Talami R, Pei W, **Miller C**, Stouffs R. *A framework for building material recycling potential: Whole-life carbon assessment for residential buildings at city-scale*. J Phys Conf Ser. 2025;3140(13):132008. <https://doi.org/10.1088/1742-6596/3140/13/132008>
11. Malakhataka E*, Lodder T, Teli D, Sindelar D, **Miller C**, Wallbaum H. *Insights into personalized thermal comfort through innovative data collection: Case study from HSB Living Lab*. J Phys Conf Ser. 2025;3140(7):072016. <https://doi.org/10.1088/1742-6596/3140/7/072016>
12. Gottkehas Kamp BG*, Ignatius M, Lim J, Fujiwara K, Hepf C, Biljecki F, **Miller C**. *Walking the heat: why thermal walks matter for high resolution microclimate mapping*. J Phys Conf Ser. 2025;3140(8):082001. <https://doi.org/10.1088/1742-6596/3140/8/082001>
13. You J*, Gottkehas Kamp BG, Stouffs R, Frei M, Wang D, **Miller C**. *UTCI vs. human perception: assessing thermal comfort for the aging population*. J Phys Conf Ser. 2025;3140(8):082007. <https://doi.org/10.1088/1742-6596/3140/19/082007>
14. **Miller C***, Ibrahim M, Akbar IS, Picchetti B, Chua YX, Frei M, Biljecki F, Chong A, Quintana M, Fu C. *The Cool, Quiet City machine learning competition: Overview and results*. J Phys Conf Ser. 2025;3140(11):112017. <https://doi.org/10.1088/1742-6596/3140/11/112017>
15. **Miller C***, Chua YX, Frei M, Zhang W, Kyaw GMM, Seah JX, Fan XJ, Parkinson T, Zhang H, Lo JC, Lee J, Schiavon S. *Wrist to Rest: A pilot study to characterize sleep habits and bedroom environments using scalable watch-based microsurveys*. J Phys Conf Ser. 2025;3140(7):072005. <https://doi.org/10.1088/1742-6596/3140/7/072005>
16. Wang D*, Lim J, Ignatius M, Fujiwara K, Gottkehas Kamp B, Biljecki F, Wong NH, **Miller C**. *Enhancing urban digital twin interfaces to support thermal comfort planning*. J Phys Conf Ser. 2025;3140(8):082017. <https://doi.org/10.1088/1742-6596/3140/20/082017>
17. Fan X, Zhang W, Mani R, **Miller C**, Parkinson T, Lo JC, Lee JK, Zhang H, Schiavon S*. *Bedroom ventilation and air quality during sleep: Insights from a pilot field study in Singapore*. Healthy Buildings Europe 2025 - Proceedings of an ISIAQ International Conference; 2025:130-137.
18. Chan JH*, You J, Stouffs R, Gottkehas Kamp BG, **Miller C**. *A multi-modal wearable and machine learning framework for accurate indoor-outdoor classification in urban climate studies*. IOP Conf Ser Earth Environ Sci. 2025;1554(1):012060. <https://doi.org/10.1088/1755-1315/1554/1/012060>
19. Ignatius M, Tong S, Lim J, Xu R, Lu Y, Tan E, Biljecki F, **Miller C**, Wong NH*. *Campus as a Living Lab: Using the BEAM Framework and Digital Twin to Address Urban Microclimate Challenges*. 12th International Conference on Urban Climate, Rotterdam, The Netherlands, 7–11 Jul 2025, ICUC12-867, <https://doi.org/10.5194/icuc12-867,2025>
20. Mosteiro-Romero M*, Park Y, **Miller C**. *People in Cities: Combining subjective occupant feedback with urban-scale data to support indoor and outdoor thermal comfort*. ASIM '24: Proceedings of the 5th Asia Conference of the International Building Performance Simulation Association 2024; 2024 pp. 277-284. https://publications.ibpsa.org/proceedings/asim/2024/papers/A18_asim2024_1183.pdf
21. Zhang Y, Favero M, Chwalek P, Zhong S, Lalanne D, Paradiso J, **Miller C**, Sonta A*. *Mind the hazard: Modeling and interpreting comfort with personalized sensing*. BuildSys '24: Proceedings of the 11th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation. ACM; 2024:209-213. <https://doi.org/10.1145/3671127.3698188>
22. Ignatius M, Lim J, Gottkehas Kamp B, Fujiwara K, **Miller C**, Biljecki F*. *Digital twin and wearables unveiling pedestrian comfort dynamics and walkability in cities*. ISPRS Ann Photogramm Remote Sens Spat Inf Sci. 2024;X-4/W5-2024: 195–202. <https://doi.org/10.5194/isprs-annals-x-4-w5-2024-195-2024>
23. Mosteiro-Romero M*, Park Y, **Miller C**. *Converging Smartwatch and Urban Datasets for Sustainable City Planning: A Case Study in Seoul, South Korea*. BuildSim Nordic 2024 - International Building Performance Simulation Association Conference. 2024. <https://doi.org/10.1051/e3sconf/202456203004>
24. **Miller C***, Tan C. *Data science skills for the built environment: Lessons learned from a massive open online Python course for construction, architecture, and engineering*. BuildSim Nordic 2024 - International Building Performance Simulation Association Conference. 2024. <https://doi.org/10.1051/e3sconf/202456206001>
25. Fu C, Kazmi H, Quintana M, **Miller C***. *Enhancing Classification of Energy Meters with Limited Labels using a Semi-Supervised Generative Model*. Proceedings of the 10th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation. Association for Computing Machinery; 2023. pp. 450–453. <https://doi.org/10.1145/3600100.3626633>

26. Mosteiro-Romero M*, Quintana M, **Miller C**, Stouffs R. *From Personal Comfort to District Performance: Using Smartwatch and WiFi Data for Occupant-Driven Operation*. BuildSys '23: Proceedings of the 10th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation. Association for Computing Machinery; 2023. pp. 278–279. <https://doi.org/10.1145/3600100.3626259>
27. Ramani V, Ignatius M, Lim J, Biljecki F, **Miller C***. *A Dynamic Urban Digital Twin Integrating Longitudinal Thermal Imagery for Microclimate Studies*. BuildSys '23: Proceedings of the 10th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation. Association for Computing Machinery; 2023. pp. 421–428. <https://doi.org/10.1145/3600100.3626345>
28. **Miller C***, Quintana M, Frei M, Chua YX, Fu C, Picchetti B, Yap W, Chong A, Biljecki F. *Introducing the Cool, Quiet City Competition: Predicting Smartwatch-Reported Heat and Noise with Digital Twin Metrics*. BuildSys '23: Proceedings of the 10th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation. Association for Computing Machinery; 2023. pp. 298–299. <https://doi.org/10.1145/3600100.3626269>
29. Maisha K, Frei M, Quintana M, Chua YX, Jain R, **Miller C***. *Utilizing wearable technology to characterize and facilitate occupant collaborations in flexible workspaces*. J Phys Conf Ser. 2023;2600: 142009. <https://doi.org/10.1088/1742-6596/2600/14/142009>
30. Jin X, Fu C, Kazmi H, Balint A, Canaydin A, Quintana M, Biljecki F, Xiao F, **Miller C***. *The Building Data Genome Directory – An open, comprehensive data sharing platform for building performance research*. J Phys Conf Ser. 2023;2600: 032003. <https://doi.org/10.1088/1742-6596/2600/3/032003>
31. Rysanek A*, Nagy Z, **Miller C**, Dilsiz AD. *How good is the advice from ChatGPT for building science? Comparison of four scenarios*. J Phys Conf Ser. 2023;2600: 082006. <https://doi.org/10.1088/1742-6596/2600/8/082006>
32. Mosteiro-Romero M*, **Miller C**, Quintana M, Chong A, Stouffs R. *Leveraging campus-scale Wi-Fi data for activity-based occupant modeling in urban energy applications*. J Phys Conf Ser. 2023;2600: 132008. <https://doi.org/10.1088/1742-6596/2600/13/132008>
33. Tartarini F, Frei M, Schiavon S, Chua YX, **Miller C***. *Cozie Apple: An iOS mobile and smartwatch application for environmental quality satisfaction and physiological data collection*. J Phys Conf Ser. 2023;2600: 142003. <https://doi.org/10.1088/1742-6596/2600/14/142003>
34. **Miller C***, Christensen R, Leong JK, Abdelrahman M, Tartarini F, Quintana M, Mueller AM, Frei M. *Smartwatch-based ecological momentary assessments for occupant wellness and privacy in buildings*. Indoor Air 2022 - 17th International Conference of the International Society of Indoor Air Quality & Climate. 2022. <http://arxiv.org/abs/2208.06080>
35. **Miller C***, Chua YX, Frei M, Quintana M. *Towards smartwatch-driven just-in-time adaptive interventions (JITAI) for building occupants*. BuildSys '22: The 9th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys '22), Nov. 9–10, 2022, Boston, MA, USA. Association for Computing Machinery. 2022. <https://doi.org/10.1145/3563357.3566135>
36. Fu C, Arjunan P, **Miller C***. *Trimming outliers using trees: Winning solution of the Large-scale Energy Anomaly Detection (LEAD) competition*. BuildSys '22: The 9th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys '22), Nov. 9–10, 2022, Boston, MA, USA. Association for Computing Machinery. 2022. <https://doi.org/10.1145/3563357.3566147>
37. Quintana M, Nagy Z, Tartarini F, Schiavon S, **Miller C***. *ComfortLearn: Enabling agent-based occupant-centric building controls*. RLEM '22: Third ACM SIGEnergy Workshop on Reinforcement Learning for Energy Management in Buildings & Cities (RLEM), Nov. 9–10, 2022, Boston, MA, USA. Association for Computing Machinery. 2022. <https://doi.org/10.1145/3563357.3566167>
38. Teo YT, Quintana M, Sabarudin MZB, Tan C, Chong A, **Miller C***. *Dataset: Green Mark certified buildings metadata from Singapore*. DATA '22: In The Fifth International Workshop on Data: Acquisition To Analysis, Nov. 6–9, 2022, Boston, MA, USA. Association for Computing Machinery. 2022. <https://doi.org/10.1145/3560905.3567771>
39. Leprince J*, **Miller C**, Madsen H, Basu K, van der Vlist R, Zeiler W. *Grey-brick buildings, an open data set of calibrated RC models of Dutch residential building heat dynamics*. DATA '22: The Fifth International Workshop on Data: Acquisition To Analysis, Nov. 6–9, 2022, Boston, MA, USA. Association for Computing Machinery. 2022. <https://doi.org/10.1145/3560905.3567760>
40. Zhan S, Quintana M, **Miller C**, Chong A*. *From model-centric to data-centric: A practical MPC implementation framework for buildings*. BuildSys '22: The 9th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation, Nov. 9–10, 2022, Boston, MA, USA. Association for Computing Machinery. 2022. <https://doi.org/10.1145/3563357.3564077>

41. **Miller C***, Hao L, Fu C. *Gradient boosting machines and careful pre-processing work best: ASHRAE Great Energy Predictor III lessons learned*. ASHRAE Annual Conference 2022. 2022. <http://arxiv.org/abs/2202.02898>
42. Naveen Mani Kumar B, Anirudh ME, Jeevakaarthik D, **Miller C**, Biljecki F*. *Thermal image analysis of Singapore's housing infrastructure*. IRC-SET 2021. Springer Nature Singapore. 2022. pp. 163–176. https://doi.org/10.1007/978-981-16-9869-9_13
43. Mosteiro-Romero M*, Alva P, **Miller C**, Stouffs R. *Towards occupant-driven district energy system operation: A digital twin platform for energy resilience and occupant well-being*. Proceedings of the 36th International Conference on Passive and Low Energy Architecture, PLEA Santiago 2022. 2022. <https://bit.ly/3FiroD3>
44. Alva P, Mosteiro-Romero M, **Miller C**, Stouffs R*. *Digital twin-based resilience evaluation of district-scale archetypes: A COVID-19 scenario case study using a university campus pilot*. POST-CARBON, Proceedings of the 27th International Conference of the Association for Computer Aided Architectural Design Research in Asia (CAADRIA) 2022, Sydney, Australia. 2022. p. 1 : 525. <https://bit.ly/3Ziiu0k>
45. Leprince J*, **Miller C**, Frei M, Madsen H, Zeiler W. *Fifty shades of black: uncovering physical models from symbolic regressions for scalable building heat dynamics identification*. BuildSys '21: The 8th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation, Nov. 17–18, 2021, Coimbra, Portugal. Association for Computing Machinery. 2021. pp. 345–348. <http://dx.doi.org/10.1145/3486611.3491120>
46. Arjunan P*, Dobler G, Lee K, **Miller C**, Biljecki F, Poolla K. *Operational characteristics of residential air conditioners with temporally granular remote thermographic imaging*. BuildSys '21: The 8th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation, Nov. 17–18, 2021, Coimbra, Portugal. ACM, Association for Computing Machinery. 2021. pp. 184–187. <http://dx.doi.org/10.1145/3486611.3486659>
47. Quintana M, Abdelrahman M, Frei M, Tartarini F, **Miller C***. *Longitudinal personal thermal comfort preference data in the wild*. SenSys '21: The 19th ACM Conference on Embedded Networked Sensor Systems, Nov. 15–17, 2021, Coimbra, Portugal. ACM, Association for Computing Machinery. 2021. pp. 556–559. <http://dx.doi.org/10.1145/3485730.3493693>
48. **Miller C***, Abdelrahman M, Chong A, Biljecki F, Quintana M, Frei M, Chew M, Wong D. *The Internet-of-Buildings (IoB) — Digital twin convergence of wearable and IoT data with GIS/BIM*. J Phys Conf Ser. 2021;2042: 012041. <http://dx.doi.org/10.1088/1742-6596/2042/1/012041>
49. Quintana M, Schiavon S, Tham KW, **Miller C***. *Balancing thermal comfort datasets: We GAN, but should we?*. BuildSys '20: Proceedings of the 7th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation. Association for Computing Machinery. 2020. pp. 120–129. <http://dx.doi.org/10.1145/3408308.3427612>
50. Sae-Zhang P, Quintana M, **Miller C***. *Differences in thermal comfort state transitional time among comfort preference groups*. 16th Conference of the International Society of Indoor Air Quality and Climate: Creative and Smart Solutions for Better Built Environments, Indoor Air 2020. 2020. p. 166587. <https://bit.ly/3J4vodH>
51. Abdelrahman M, Chong A, **Miller C***. *Build2Vec: Building Representation in Vector Space*. SimAUD Conference Online. 2020. <http://arxiv.org/abs/2007.00740>
52. Jayathissa P*, Quintana M, Sood T, Nazarian N, **Miller C**. *Is your clock-face cozie? A smartwatch methodology for the in-situ collection of occupant comfort data*. J Phys Conf Ser. 2019; 1343: 012145. <http://dx.doi.org/10.1088/1742-6596/1343/1/012145>
53. Sood T*, Quintana M, Jayathissa P, Abdelrahman M, **Miller C**. *The SDE4 Learning Trail: Crowdsourcing occupant comfort feedback at a net-zero energy building*. J Phys Conf Ser. 2019; 1343: 012141. <http://dx.doi.org/10.1088/1742-6596/1343/1/012141>
54. Quintana M, **Miller C***. *Towards Class-Balancing Human Comfort Datasets with GANs*. BuildSys '19: Proceedings of the 6th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation. Association for Computing Machinery. 2019. pp. 391–392. <http://dx.doi.org/10.1145/3360322.3361016>
55. **Miller C***, Quintana M, Glazer J. *Twenty years of building performance analysis trends: A topic modeling analysis of the Bldg-Sim email list archive*. Proceedings of Building Simulation 2019: 16th Conference of IBPSA. 2019. <http://dx.doi.org/10.26868/25222708.2019.211087>
56. Park JY, **Miller C**, Nagy Z*. *A Data-Driven Load Shape Profile Based Building Benchmarking: Comparing DOE Reference Buildings With A Large Metering Dataset*. Proceedings of Building Simulation 2019: 16th Conference of IBPSA. 2019. <http://dx.doi.org/10.26868/25222708.2019.211074>

57. **Miller C***. *Predicting success of energy savings interventions and industry type using smart meter and retrofit data from thousands of non-residential buildings*. BuildSys '17: Proceedings of the 4th ACM International Conference on Systems for Energy-Efficient Built Environments. Association for Computing Machinery. 2017. pp. 1–4. <http://dx.doi.org/10.1145/3137133.3137160>
58. **Miller C***, Meggers F. *The Building Data Genome Project: An open, public data set from non-residential building electrical meters*. Energy Procedia. 2017; 122: 439–444. <http://dx.doi.org/10.1016/j.egypro.2017.07.400>
59. **Miller C***, Hersberger C, Jones M. *Automation of common building energy simulation workflows using Python*. Proceedings of BS2013: 13th International Conference of the International Building Performance Simulation Association, IBPSA. IBPSA; 2013. pp. 210–217. <http://dx.doi.org/10.13140/RG.2.1.1999.4087>
60. Kalluri B*, **Miller C**, Seshadri B, Schlueter A. *A Cyber-physical middleware platform for buildings in smart cities*. Advances in Informatics and Computing in Civil and Construction Engineering. Springer International Publishing; 2019. pp. 645–652. http://dx.doi.org/10.1007/978-3-030-00220-6_77
61. Luerssen C*, Wahed A, Reindl T, **Miller C**, Cheong D, Sekhar C. *Energy storage for PV-driven air-conditioning for an off-grid resort—a case study*. ISES solar world congress 2017 and IEA SHC solar heating and cooling conference for buildings and industry. 2017. pp. 1785–1795. <http://dx.doi.org/10.18086/swc.2017.28.14>
62. Fonseca JA*, **Miller C**, Schlueter A. *Unsupervised load shape clustering for urban building performance assessment*. Energy Procedia. 2017; 122: 229–234. <http://dx.doi.org/10.1016/j.egypro.2017.07.350>
63. Rysanek A*, **Miller C**, Schlueter A. *A workflow for managing building information and performance data using virtual reality: an alternative to BIM for existing buildings?* Building Simulation 2017. 2017. <https://doi.org/10.26868/25222708.2017.817>
64. **Miller C***, Schlueter A. *Forensically discovering simulation feedback knowledge from a campus energy information system*. Proceedings of the Symposium on Simulation for Architecture & Urban Design. San Diego, CA, USA: Society for Computer Simulation International; 2015. pp. 136–143. <https://dl.acm.org/doi/10.5555/2873021.2873040>
65. Rysanek AM*, Murray PJ, Pantelic J, **Miller C**, Meggers F, Schlueter A. *The design of a decentralized ventilation system for an office in Singapore: key findings for future research*. Proceedings of International Conference CISBAT 2015 Future Buildings and Districts Sustainability from Nano to Urban Scale. LESO-PB, EPFL; 2015. pp. 77–82. <https://doi.org/10.5075/epfl-cisbat2015-77-82>
66. **Miller C***, Thomas D, Kämpf J, Schlueter A. *Long wave radiation exchange for urban scale modelling within a co-simulation environment*. Proceedings of International Conference CISBAT 2015 Future Buildings and Districts Sustainability from Nano to Urban Scale. LESO-PB, EPFL; 2015. pp. 871–876. <http://dx.doi.org/10.5075/epfl-cisbat2015-871-876>
67. **Miller C***, Thomas D, Irigoyen SD, Hersberger C, Nagy Z, Rossi D, Schlueter A. *BIM-extracted EnergyPlus model calibration for retrofit analysis of a historically listed building in Switzerland*. Proceedings of SimBuild 2014: 2014 ASHRAE/IBPSA-USA Building Simulation Conference, ASHRAE/IBPSA. 2014. https://publications.ibpsa.org/proceedings/simbuild/2014/papers/simbuild2014_42.pdf
68. **Miller C***, Nagy Z, Schlueter A. *A seed dataset for a public, temporal data repository for energy informatics research on commercial building performance*. In proceeding of: 3rd Conf on Future Energy Business & Energy Informatics, Rotterdam, Netherlands. 2014. pp. 1–2. <http://dx.doi.org/10.13140/RG.2.1.4620.8485>
69. Thomas D*, **Miller C**, Kampf J, Schlueter A. *Multiscale co-simulation of EnergyPlus and CitySim models derived from a building information model*. Bausim 2014: Fifth German-Austrian IBPSA Conference. 2014. pp. 469–476. <http://dx.doi.org/10.13140/2.1.4639.1040>
70. **Miller C***, Schlueter A. *Applicability of lean production principles to performance analysis across the life cycle phases of buildings*. Proceedings of the CLIMA 2013: 11th REHVA Congress and 8th International Conference on IAQVEC, REHVA. 2013. <http://dx.doi.org/10.13140/RG.2.1.3572.2729>
71. **Miller C***, Sekhar C. *Development of a dual duct air terminal unit in EnergyPlus to model centralized dedicated outdoor air systems (DOAS) in cooling mode*. Proceedings of IAQVEC 2010: The 7th International Conference on Indoor Air Quality, Ventilation, and Energy Conservation in Buildings. 2010. <http://dx.doi.org/10.13140/RG.2.1.3965.4884>
72. **Miller C***, Sekhar C. *Modeling of the single coil, twin fan air-conditioning system in EnergyPlus*. Proceedings of the Fourth National Conference of IBPSA-USA: SimBuild 2010. 2010;4: 201–206. https://publications.ibpsa.org/proceedings/simbuild/2010/papers/simbuild2010_TS04B-03-Miller.pdf

Book Chapters

1. Gunay B, Hobson B, Ouf M, Nagy Z, **Miller C**. *Design of sequences of operation for occupant-centric controls*. In: O'Brien W, Tahmasebi F, editors. *Occupant-Centric Simulation-Aided Building Design Theory, Application, and Case Studies*. Routledge; 2023. <https://bit.ly/3W9r0g0>

Editorials

1. **Miller C**, Park JY, Kazmi H. *Editorial: Message from General and Technical Program Chairs*. BuildSys 2026 - Proceedings of the 13th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation, pp. III–IV. <https://dl.acm.org/doi/proceedings/10.1145/3744256>
2. Ardakanian O, **Miller C**. *Editorial: Message from General Chairs*. ACM Sustainability Week Companion 2026 - Proceedings of the 2026 ACM Sustainability Week, pp. III–IV. <https://dl.acm.org/doi/proceedings/10.1145/3765611>
3. Nagy, Z, **Miller C**, Gupta, R. *Editorial: Message from General and Technical Program Chairs*. BuildSys 2023 - Proceedings of the 10th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation, pp. III–IV. <https://dl.acm.org/doi/proceedings/10.1145/3600100>
4. Salamone F, Chinazzo G, **Miller C**, Sibilio S, Masullo M. *Editorial: Innovative Human-Centric Investigations and Technologies for Human Wellbeing and Health in the Built Environment*. *Frontiers in Built Environment*. 2022;8. <http://dx.doi.org/10.3389/fbuil.2022.918643>

Research Funding

1. **PI**, *HEATS: Heat Exposure, AcTivity, and Sleep – Heat Exposure and Nudging Sub-Projects*, 2024-2026, Singapore National Research Foundation (NRF) CREATE Thematic Grant Call
2. **Co-PI**, *HEATS: Heat Exposure, AcTivity, and Sleep – Cooling Homes and Cooling Dorms Sub-Projects*, 2024-2026, Singapore National Research Foundation (NRF) CREATE Thematic Grant Call
3. **Co-PI**, *Multi-scale Digital Twins for the Urban Environment: From Heartbeats to Cities - Phase 2*, 2023-2024, Reimagine Research Scheme Grant through the NUS Office of the Deputy President of Research and Technology (ODPRT)
4. **PI**, *Towards a BRICK Translation and Mapping Engine with Applications: Automating Building Semantic Data for Scalable Plug-n-Play Onboarding and Analytics*, 2021-2024, Johnson Controls Open Blue Innovation Center EDB Collaborative Project
5. **Co-PI**, *Supporting Cooling NUS Project with BEAM Initiative: Baselining, Evaluating, Action, And Monitoring NUS Kent Ridge Campus*, 2022-2027, NUS University Campus Infrastructure (UCI)
6. **Travel Grant Recipient**, *Developing a digitalised user-centric POE process that monitors and predicts long-term users' energy use and comfort behaviour void of accuracy-based errors*, 2023, Massey University International Visitor Research Fund 2023
7. **Co-PI**, *Multi-scale Digital Twins for the Urban Environment: From Heartbeats to Cities - Phase 1*, 2022-2023, NUS Reimagine Research Scheme through the Office of the Deputy President of Research and Technology
8. **Co-PI**, *Mixed-mode Ventilation for Office Buildings in the Tropics*, 2021-2023, Cities of Tomorrow (CoT) Grant from the Ministry of National Development (MND) through the Building and Construction Authority (BCA)
9. **PI**, *Ecological Momentary Assessment (EMA) for Built Environment Research*, 2020-2023, Ministry of Education (MOE) Tier 1 Grant through the School of Design and Environment
10. **PI**, *Buildings Can Influence My Health? Using SDE Well Buildings, Wearables, and Data Science to Learn about Buildings' Impact on Occupant Well Being*, 2020-2023, Learning Innovation Fund - Technology (LIFT) Grant
11. **Co-PI**, *The Internet-of-Buildings (IoB) Platform – Visual Analytics for AI Technologies towards a Well and Green Built Environment*, 2020-2023, Ministry of Education (MOE) Tier 1 Grant through the NUS School of Design and Environment
12. **Co-PI**, *Future Resilient Systems (FRS) Module 1.4 - Digital Twin-Enabled System Resilience* NUS Collaboration with the Future Resilient Systems Laboratory 2, Singapore-ETH Center (SEC), 2020-2025, Singapore National Research Foundation (NRF)
13. **PI**, *Blended Module Development - PF1103*, 2019-2021, Learning Innovation Fund - Technology (LIFT) Grant

14. **PI**, *edX Course Development - Intro to Data Science for the Built Environment*, 2019-2021, Learning Innovation Fund - Technology (LIFT) Grant
15. **PI**, *Spacematch: AI-Enhanced Spatial Recommendation System for Co-Working and Activity-Based Working Spaces*, 2019, NUS Technology Acceleration Program (TAP)
16. **Technical Team Lead**, *ASHRAE Great Energy Predictor III Competition Prize Funding*, 2019, ASHRAE Organization
17. **PI**, *Temporal Mining of Energy and Indoor Environmental Quality Data from Buildings*, 2018-2021, NUS Start-up Grant
18. **PI and Theme D Co-leader**, *Building Efficiency and Sustainability in the Tropics 2 (SinBerBEST2) Lab - Theme D Data Analytics for the Built Environment* NUS Collaboration with the Berkeley Alliance for Research in Singapore (BEARS), 2017-2023, Singapore National Research Foundation (NRF)
19. **PI**, *Using Social Comparison to Promote Electricity and Water Efficiency in Singapore* (NUS Sub-Project), 2017-2020, Part of the BSEWWT (Behavioural Studies in the Energy, Water, Waste, and Transport) Grant from the Singapore National Research Foundation (NRF) through the Future Resilient Systems (FRS) Laboratory at the Singapore ETH Centre (SEC)
20. **PI**, *Smart Devices & Energy Savings in NUS Project*, 2018-2019, (Co-PI from 2017-2018), NUS Office of the Deputy President of Research and Technology (ODPRT)

Highlighted Major Keynote Presentations

1. **Brains4Buildings Final Conference**, Invited Keynote Speaker with Travel Expenses Sponsored - Delft, Netherlands - *AI is not Magic: Understanding Capabilities While Cutting Through the Hype* - Oct. 2025
2. **BuildSim Nordic 2024, The Biennial Conference of the International Building Performance Simulation Association (IBPSA) - Nordic**, Invited Keynote Speaker with Travel Expenses Sponsored - Helsinki, Finland - *How AI Disrupts: Lessons for the Built Environment from Other Industries* - Jun. 2024
3. **AI in AEC 2024 Conference - Artificial Intelligence in Architecture, Engineering and Construction** - Invited Keynote Speaker with Travel Expenses Sponsored - Helsinki, Finland - *How AI Disrupts: Lessons for the Built Environment from Other Industries* - Mar. 2024
4. **CLIMA 2022, The 14th REHVA HVAC World Congress**, Digitization Theme - Invited Keynote Speaker with Travel Expenses Sponsored - Rotterdam, Netherlands - *Scalability, Benchmarking and Skills: A Roadmap for Built Environment Digitization* - May 2022
5. **eSim 2021, The Biennial Conference of International Building Performance Simulation Association (IBPSA) - Canada**, Invited Keynote Speaker - Online - *Measured data? Simulated data? What's the future? Bridging the gap between physics-based and data-driven modeling* - Jun. 2021
6. **BuildSim Nordic 2020, The Biennial Conference of the International Building Performance Simulation Association (IBPSA) - Nordic**, Invited Keynote Speaker - Online - *The Great Energy Predictor III Kaggle Competition - How can we bridge physics-based and data-driven modeling?* - Oct. 2020

Externally-sponsored/Financially-supported Invited Research Talks

1. **Chalmers University - Data for a Sustainable Built Environment Seminar** - Invited Speaker with Sponsored Travel Expenses (with Doctoral Defense Examination duties) - Gothenburg, Sweden - *Crowdsourcing Data and Models for AI-driven Urban Heat and Energy Solutions* - Jan. 2026
2. **MuSIC Doctoral Network Training Week**, Invited Speaker with Sponsored Travel Expenses - Nicosia, Cyprus - *Field-Based Human-Centric Research: Lessons Learned for PhD Students* - May 2025
3. **KU Leuven, Ghent Campus - Meet the Jury Seminar** - Invited Speaker with Sponsored Travel Expenses (with Doctoral Defense Examination duties) - Ghent, Belgium - *The Human Dimension: Using wearable and IoT data to understand buildings and cities better* - Mar. 2025
4. **ISHRAE TechCon 2025 - High Performance Building & Critical Infrastructures** - Invited Speaker with Sponsored Travel Expenses - Manama, Bahrain and Doha, Qatar - *Sustainable Buildings in Hot & Humid Regions: A Human-Centric Perspective* - Jan. 2025
5. **5th International Summer School on Data Science and AI for Energy Engineers** - Invited Speaker with Sponsored Travel Expenses - Leuven, Belgium - *Exploring the Human Dimensions of Energy Forecasting through Open Data and Models* - Jul. 2024

6. **13th International Conference on Engineering, Project, and Production Management (EPPM) 2023** - Invited Speaker with Sponsored Travel Expenses through a New Zealand Academic Travel Grant - Auckland, New Zealand - *Data Science and Big Data Skills for the Built Environment* - Nov. 2023
7. **Asian Development Bank (ADB) NUS Programme for Digital Transformation - Towards a Smart Nation**, Invited Speaker/Panelist with Honorarium - Singapore - *The Human Dimension - Using wearable and IoT data to better understand cities* - Sep. 2022
8. **Aalto University Post-graduate Seminar on Data-Driven Smart Buildings**, Invited Speaker with Sponsored Travel Expenses - Espoo, Finland - *The Role of Open Data and Benchmarking Machine Learning for Built Environment Applications* - Sep. 2022
9. **Tsinghua University - Department of Building Science Seminar**, Invited Speaker with Honorarium - Online - *More Open Data/Models and Better Skills: The Role of Open Data, Machine Learning Skills, and Crowdsourced Models for the Built Environment* - Feb. 2022
10. **C3.ai Digital Transformation Institute - Digital Transformation of the Built Environment Workshop**, Invited Speaker with Honorarium - Online - *The Role of Machine Learning Skills and Crowdsourced Models for the Built Environment* - Oct. 2021
11. **Architectonics Symposium, ASHRAE and ISHRAE Mumbai Chapter**, Invited Speaker (Master Speaker) with Sponsored Travel Expenses - Mumbai, India - *The SDE4 NZEB Case Study: The First New Construction Net Zero Energy (NZEB) and WELL Certified Building in Singapore* - Jul. 2019
12. **International Building Performance Simulation Association (IBPSA) Asia (ASIM) Conference 2018**, Invited Speaker with Sponsored Conference Registration - Hong Kong SAR - *The Importance of Open Source Data for Built Environment Machine Learning Research* - Dec. 2018
13. **Croucher Advanced Study Institute - Symposium on Smart Building Technology, Management, and Fog Computing**, Invited Speaker with Sponsored Travel Expenses - Hong Kong SAR - *Impacting People's Lives Through Data-Driven Innovation in the Built Environment* - Jan. 2018

Invited Panel Participant/Moderator and Minor Keynote Talks

1. **ACM BuildSys 2026 Panel**, Panel Planner and Moderator - Banff, Canada - *Big Questions About AI* - Jun. 2026
2. **Newcastle University Conference on AI and Data Analytics for Infrastructure, Buildings and Communities Panel**, Invited Panelist - Singapore - *A Like Button for Spaces: Capturing City-Scale Human Experiences via Watch-based Microsurveys* - Apr. 2025
3. **buildingSMART International Summit - Singapore 2024 Spin-off Event: The Role of BIM in Implementing Smart Buildings**, Invited Panelist - Singapore - *Translating the Languages of Buildings: The Importance of System Interoperability* - Mar. 2025
4. **ASHRAE Singapore Distinguished Lecture**, Invited Speaker - Singapore - *A Like Button for Spaces: Capturing City-Scale Human Experiences via Watch-based Microsurveys* - Mar. 2025
5. **Norway-Singapore Science Week - Workshop: Sustainable Cities and Neighbourhoods**, Invited Panelist and Speaker - Singapore - *Human-Building Conversations - Better ways to detect how people perceive the built environment* - Sep. 2023
6. **The 11th Energy Management Action Network Workshop - EMAK11 - Toward Net-Zero Energy Building** - Invited Panelist (Keynote) and Speaker - Singapore - *The Human Dimension of Net Zero Energy Buildings* - Feb. 2023
7. **Norway-Singapore: Smart Nations Workshop - Session 1: Smart and Sustainable Urban Development**, Invited Panelist and Speaker - Singapore - *Sustainable Urban Development R&D in Singapore - Human-centric and Digital Twin-based Approaches* - Jan. 2023
8. **Norway-Singapore Science Week - Workshop: Sustainable Cities and Neighbourhoods**, Invited Panelist and Speaker - Singapore - *The Human Dimension - Using field-based intensive longitudinal data to characterize humans and buildings* - Nov. 2022
9. **United Architects of the Philippines (UAP) - Glasstech Asia 2022: Climate Change and the Future of Sustainability Seminar**, Invited Keynote Speaker and Panelist - Singapore - *Human-centric Machine Learning for Sustainable Buildings* - Oct. 2022
10. **National Technology Enhanced Learning (NTEL) Conference 2022- Panel on Empowering Learning in the Cyber-Physical Era**, Invited Panelist - Singapore - *Teaching Coding Remotely using a MOOC* - Oct. 2022

11. **FOSSASIA Summit - Open Science Panel**, Invited Panelist - Online - Mar. 2022
12. **Becoming an edX Prize Finalist Panel**, Invited Panelist - Online - Feb. 2022
13. **Asian Development Bank Institute Annual Conference - Climate Change Mitigation and Green Finance**, Invited Panelist - Online - *Future-proofing sustainable cooling demand* - Dec. 2021
14. **13th International Conference on Applied Energy (ICAE 2021) - Open Data for Urban Energy Systems**, Invited Panelist - Online - *More Open Data and Better Skills - The Role of Machine Learning Skills and Crowdsourced Models for the Built Environment* - Nov. 2021
15. **Norway-Singapore Science Week– Webinar Program - Sustainable Urban Development**, Invited Panelist and Moderator - Online - *SDE4 Net Zero Energy Building and Technology* - Nov. 2021
16. **Eden Strategy Institute/IMDA - Focus Group for the Tech for Sustainability Playbook**, Invited Panelist - Online - Sep. 2021
17. **International Conference on Sustainability in Energy and Buildings SEB-21**, Invited Keynote Speaker - Online - *Leveraging Open Energy Data for Buildings - From the ASHRAE Great Energy Predictor III Competition to the Annex 81 Open Data Sandbox* - Sep. 2021
18. **IEEE Power & Energy Society General Meeting - The Interplay Between Building Energy Efficiency and Demand Response: Implications for Power Systems**, Invited Panelist - Online - *The ASHRAE Great Energy Predictor III Competition: Using Kaggle Machine Learning Model Crowdsourcing for Meter Prediction Innovation* - Jul. 2021
19. **Occupant-centric Design Panel Discussion, 6th Expert Meeting of IEA EBC - Annex 79 - Occupant-Centric Building Design and Operation**, Panel Moderator and Planner - Online - *What exactly is occupant-centric building design? – Different perspectives but common sense or domain-dependent approaches?* - Apr. 2021
20. **International Facility Management Expo Series (iFAME) - Webinar: Remodeling FM for the Future**, Invited Panel Speaker - Online - *Remodeling Data Science with FM* - Feb. 2021
21. **Norway-Singapore Science Week Symposium**, Invited Panel Speaker and Moderator - Online - *The Role of Universities in Sustainable Urban Development* - Oct. 2020
22. **ArchXpo 2018, Speaker at United Architects of the Philippines (UAP) Symposium**, Panel Speaker - Singapore - *Impacting People's Lives Through Data-Driven Innovation in the Built Environment* - Oct. 2018
23. **Singapore-ETH Center (ETH Zürich) - Guest Lecture at a Career Talk Seminar**, Invited Panel Speaker - Singapore - *Getting a Tenure-Track Position at NUS* - May 2018

Conference Seminar, Webinar, and Workshop Presentations/Posters (w/o Publication)

1. **AI 4 X 2025**, Invited Speaker - Singapore - *Cozie and Kaggle: Crowdsourcing Data and Models for AI-driven Urban Heat and Energy Solutions* - Jul. 2025
2. **University of California, Berkeley - Center for the Built Environment (CBE) Industry Day**, Invited Speaker - Berkeley, CA, USA - *Make Yourself Comfortable: Nudging Heat and Noise Mitigation Using Smartwatch Interventions* - Apr. 2025
3. **Comfort At The Extremes (CATE) 2024 Conference**, Poster Presenter - Seville, Spain - *Towards scalable smartwatch-collected subjective data collection for environmental comfort at the city-scale* - Nov. 2024
4. **MuSIC Doctoral Network Training Week**, Invited Speaker - Online - *Using Wearables and IoT Data to Understand Buildings and Cities Better* - May 2024
5. **AIVC-ASC Technical Conference 2024 - Ventilation, IEQ, and Sustainability**, Invited Speaker - Singapore - *Using Wearables and IoT Data to Understand Buildings and Cities Better* - Apr. 2024
6. **Digital Transition in the Design of Built Environment Seminar - Nanjing University**, Invited Speaker - Online - *The Human Dimension - Using wearable and IoT data to better understand cities* - Dec. 2023
7. **IBPSA-USA Research Committee Webinar**, Invited Speaker - Online - *Skills for Tackling Built Environment Challenges* - Jan. 2023
8. **Fifth Plenary Meeting of IEA EBC - Annex 81 - Data-Driven Smart Buildings**, Activity Leader and Speaker - Gothenburg, Sweden - *Annex 81 - Activity D3 Update* - Oct. 2022
9. **8th International Symposium on Occupant Behaviour (OB-22) and 9th Expert Meeting of IEA EBC - Annex 79 - Occupant-Centric Building Design and Operation**, Lead Planner, Facilitator, and Speaker - Singapore - *OB-22 and Annex 79 Opening and Closing* - Sep. 2022

10. **US National Science Foundation (NSF) - International Research Experiences for Students (IRES) Seminar** - The Center of Leadership Development in Built Environment Sustainability, Invited Speaker - Singapore - *The Human Dimension - Using field-based intensive longitudinal data to characterize humans and buildings* - Sep. 2022
11. **IBPSA-Singapore and ASHRAE Singapore Chapter** - Joint Webinar, Invited Speaker - Online - *Human-Building Interaction using Wearables and ML Benchmarking* - Aug. 2022
12. **University of Witwatersrand - Construction Economics and Management (CEM) Research Series Webinar**, Invited Speaker - Online - *Scalability, Benchmarking, and Skills: A Roadmap for Built Environment Digitisation* - Jul. 2022
13. **SinBerBEST2 Program Symposium for the Berkeley Education Alliance for Research in Singapore (BEARS)** - Singapore - *InfraRed Investigation in Singapore (IRIS)* - Jul. 2022
14. **Indoor Air 2022 Conference Workshop - Workshop Planner and Presenter** - Kuopio, Finland - *Python Crash Course and Mini-Hackathon for Occupant-Centric Researchers* - Jun. 2022
15. **Vietnam Energy Efficiency Building Week VEEBW 2021 - Unlock Net-zero Commercial buildings – Rooftop Solar & Energy Efficiency Forum**, Invited Speaker - Online - *SDE4 Learning Trail and Digital Tools for NZEBs* - Nov. 2021
16. **International Facilities Management Conference 2021**, Invited Speaker - Online - *FM Powered by AI* - Nov. 2021
17. **ASHRAE Ireland Chapter Webinar**, Invited Speaker - Online - *Why Building Industry Professionals Should Learn to Code!* - Jul. 2021
18. **Singapore-IEA Regional Training Programme on Low Carbon Buildings - Module 2: Smart, Clean and Energy Efficient Building Systems and Operations**, Invited Speaker - Online - Jul. 2021
19. **IBPSA-USA Education Committee - Webinar on Programming for Energy Modellers**, Invited Speaker - Online - *Python for Building Energy Modelers* - Jul. 2021
20. **BCA-SIFMA Industry Sharing on Smart FM and Digitalisation**, Invited Speaker - Online - *Metadata Schemas for Facilities Management – Why? And How?* - May 2021
21. **7th International Symposium on Occupant Behaviour (OB-21)**, Contributing Speaker - Online - *Moving Beyond Thermal Comfort: Capturing Wellness-related Occupant Perceptions Related to Privacy, Movement and Airborne Disease Transmission* - Apr. 2021
22. **6th International Symposium on Occupant Behaviour (OB-20) and 5th Expert Meeting of IEA EBC - Annex 79 - Occupant-Centric Building Design and Operation**, Contributing Speaker and Moderator - Online - *The New Normal: Technologies to Address Rapid Digitization and Diversification of How and Where Occupants Work and Live* - Sep. 2020
23. **United Architects of the Philippines (UAP) Symposium**, Invited Speaker - Online - *Data Science for the Building Industry Research and Skills that will Change the Industry* - Sep. 2020
24. **Annex 81 Data-Driven Buildings Symposium**, Contributing Speaker - Online - *Crowdsourcing Machine Learning for Building Energy Prediction: Learning from the ASHRAE Great Energy Predictor III Kaggle Competition* - Jun. 2020
25. **ASHRAE 2020 Annual Meeting, Seminar Speaker - Seminar 49 - Occupant-Centric Building Design and Operation: Implementation Case Studies** - Online - *Spatial Optimization and Flex-Based Workspaces: A Big Opportunity for Occupant-Centric Building Controls and Operations?* - Jun. 2020
26. **ASHRAE 2020 Annual Meeting, Seminar Speaker - Seminar 12 - Winners and Winning Solutions from the ASHRAE Great Energy Predictor III Machine Learning Competition** - Online - *Overview Analysis of the Great Energy Predictor (GEP) III Competition* - Jun. 2020
27. **5th International Symposium on Occupant Behaviour (OB-20) and 4th Expert Meeting of IEA EBC - Annex 79 - Occupant-Centric Building Design and Operation**, Contributing Speaker and Moderator - Online - *What do occupants want? Let's ask them using smartwatches and Cozie* - Apr. 2020
28. **ASHRAE 2020 Winter Meeting - Seminar Speaker – Seminar 20 - Occupant-Centric Building Design and Operation: State of the Art and Challenges, Part 2** - Orlando, FL, USA - *What Do Occupants Want? Let's Ask Them* - Feb. 2020
29. **ASHRAE 2020 Winter Meeting - Seminar Speaker – Seminar 23 - The Great Energy Predictor Competition III** - Orlando, FL, USA - *Data Collection and Result Analysis for the Predictor Competition III* - Feb. 2020

30. **3rd Expert Meeting of IEA EBC - Annex 79 - Occupant-Centric Building Design and Operation**, Contributing Speaker and Moderator - Perugia, Italy - *Overview of Subtask 4 Activities and Research Introduction* - Sep. 2019
31. **SinBerBEST2 Program Symposium for the Berkeley Education Alliance for Research in Singapore (BEARS)** - Singapore - *Personalities of Buildings - Crowd-sourced Machine Learning Models for City-Scale Analytics: The Great Energy Predictor Competition 2019* - Aug. 2019
32. **ASHRAE 2019 Annual Meeting - Seminar Speaker - Seminar 62** - Kansas City, MO, USA - *The Great Energy Predictor Shootouts I and II: Revisited* - Jun. 2019
33. **US National Science Foundation (NSF) - RCN-SEES: Predictive Modeling Network for Sustainable Human-Building Ecosystems (SHBE)**, Workshop Speaker - Singapore - *Innovative Occupant Engagement: The SpaceMatch/Learning Trail Crowdsourced Feedback and Comfort Recommendation Engine* - Jan. 2019
34. **10th International Conference on Urban Climate (ICUC)**, Seminar Speaker - New York City, NY, USA - *Crowdsourcing Personal Thermal Comfort and Heat Stress Responses using Wearable Devices* - Aug. 2018
35. **SinBerBEST2 Program Symposium for the Berkeley Education Alliance for Research in Singapore (BEARS)**, Symposium Speaker - Singapore - *Personalities of Buildings - AI-Generated Building Behaviour Analysis as the Next Generation of Benchmarking* - Jul. 2018

Invited/Guest Research Talks/Visits at Universities/Organizations

1. **University of Waterloo** - Waterloo Singapore Sustainability Visit, Guest Lecture - Singapore - *Crowd-sourcing Data and Models for AI-driven Urban Solutions* - Apr. 2026
2. **Delft University of Technology (TU Delft)** - AE+T Bites Lunch Seminar, Invited Speaker - Delft, Netherlands - *The Human Dimension: Using wearable and IoT data to understand buildings and cities better* - Mar. 2025
3. **Oslo Metropolitan University (OsloMet)** - Department of Built Environment, Invited Speaker - Oslo, Norway - *Innovations in Human-Building Interaction and Modeling - Better ways to detect and predict how people perceive the built environment* - Mar. 2024
4. **Norwegian University of Science and Technology (NTNU)** - Department of Architecture and Technology, Invited Speaker - Trondheim, Norway - *Innovations in Human-Building Interaction and Modeling - Better ways to detect and predict how people perceive the built environment* - Mar. 2024
5. **Swiss Federal Institute of Technology - Lausanne (EPFL)** - Human-Oriented Built Environment Lab, Invited Speaker - Fribourg, Switzerland - *Human-Building Conversations - Better ways to detect how people perceive the built environment* - Sep. 2023
6. **Massey University** - School of the Built Environment, Invited Speaker - Auckland, New Zealand - *Human-Building Conversations - Better ways to detect how people perceive the built environment* - Aug. 2023
7. **Technical University of Denmark (DTU)** - International Centre for Indoor Environment and Energy (ICIEE) Seminar, Invited Speaker - Lyngby, Denmark - *The Human Dimension: Using field-based intensive longitudinal data to characterize humans and buildings* - Oct. 2022
8. **Stanford University** - Urban Informatics Lab Seminar, Invited Speaker (Co-invited with Filip Biljecki) - Palo Alto, CA, USA - *The Human Dimension: Using field-based intensive longitudinal data to characterize humans and buildings* - Jul. 2022
9. **University of California, Berkeley** - Center for the Built Environment Seminar, Invited Speaker (Co-invited with Adrian Chong and Filip Biljecki) - Berkeley, CA, USA - *The Human Dimension: Using field-based intensive longitudinal data to characterize humans and buildings* - Jul. 2022
10. **Lawrence Berkeley National Lab (LBNL)** - Seminar, Invited Speaker (Co-invited with Adrian Chong and Filip Biljecki) - Berkeley, CA, USA - *The Human Dimension: Using field-based intensive longitudinal data to characterize humans and buildings* - Jul. 2022
11. **University of Colorado, Boulder - National Renewable Energy Lab (NREL)** - Seminar, Invited Speaker (Co-invited with Adrian Chong and Filip Biljecki) - Boulder, CO, USA - *The Human Dimension: Using field-based intensive longitudinal data to characterize humans and buildings* - Jul. 2022
12. **Carnegie Mellon University (CMU)** - Intelligent Workplace Lab Seminar, Invited Speaker (Co-invited with Adrian Chong and Filip Biljecki) - Pittsburgh, PA, USA - *The Human Dimension: Using field-based intensive longitudinal data to characterize humans and buildings* - Jul. 2022

13. **Princeton University** - Cooling and Heating for Architecturally Optimized Systems (CHAOS) Lab Seminar, Invited Speaker (Co-invited with Adrian Chong and Filip Biljecki) - Princeton, NJ, USA - *The Human Dimension: Using field-based intensive longitudinal data to characterize humans and buildings* - Jul. 2022
14. **University of Pennsylvania** - Thermal Architecture Lab Seminar, Invited Speaker (Co-invited with Adrian Chong and Filip Biljecki) - Philadelphia, PA, USA (Hybrid) - *The Human Dimension: Using field-based intensive longitudinal data to characterize humans and buildings* - Jul. 2022
15. **Massachusetts Institute of Technology (MIT)** - Sustainable Design Lab Visit (Co-presented with Adrian Chong and Filip Biljecki) - Cambridge, MA, USA (Hybrid) - *The Human Dimension: Using field-based intensive longitudinal data to characterize humans and buildings* - Jun. 2022
16. **University of Toronto** - Building Energy and Indoor Environment Lab Seminar, Invited Speaker - Online - *The Human Dimension: Using field-based intensive longitudinal data to characterize humans and buildings* - Jun. 2022
17. **Technical Advisory Group Meeting for DOE BTO's Semantic Interoperability R&D**, Invited Participant - Online - Nov. 2021
18. **Concordia University** - Guest Research Lecture, Invited Speaker - Online - *Building and Urban Data Science Lab Overview* - Feb. 2021
19. **Swiss Federal Institute of Technology - Zurich (ETH Zürich)** - Design++ Center Guest Research Lecture, Invited Speaker - Online - *Towards Growing the Data Science Community in the Built Environment: Kaggle Competitions, MOOCs, and Encouraging Architects and Engineerings to become Coders!* - Dec. 2020
20. **University of Virginia** - Guest Research Presentation, Invited Speaker - Charlottesville, VA, USA - *Crowd-sourcing Machine Learning for Buildings: Building Data Genome Project 2.0* - Jan. 2020
21. **Princeton University** - Guest Research Meeting, Invited Speaker - Princeton, NJ, USA - *Crowdsourcing Machine Learning for Buildings: Building Data Genome Project 2.0* - Jan. 2020
22. **Kajima Technical Research Institute Singapore (KaTRIS)** - Seminar, Invited Speaker - Singapore - *Next Generation Techniques for Understanding Occupant Behavior and Satisfaction in Buildings* - Oct. 2019
23. **Lawrence Berkeley National Laboratory (LBNL)** - Lunch Talk, Invited Speaker - Berkeley, CA, USA - *Data Science Meets ASHRAE: The Great Energy Predictor Competition III* - Oct. 2019
24. **Stanford University** - Urban Informatics Lab Seminar, Invited Speaker - Palo Alto, CA, USA - *Overview of the Building and Urban Data Science Lab* - Jun. 2019
25. **Yale University** - Guest Research Lecture at School of Forestry & Environmental Studies, Invited Speaker - New Haven, CT, USA - *Personalities of Buildings - AI-Generated Building Behaviour Analysis as the Next Generation of Benchmarking* - Aug. 2018
26. **Siemens Building Technologies Corporate Headquarters** - Guest Research Lecture, Invited Speaker - Zug, Switzerland - *Towards Utilizing the Huge Amounts of Data from IoT in Buildings* - Jun. 2018
27. **Swiss Federal Institute of Technology - Zurich (ETH Zürich)** - Institute of Technology in Architecture (ITA) Guest Research Lecture, Invited Speaker - Zürich, Switzerland - *Impacting People's Lives Through Data-Driven Innovation in the Built Environment* - Jun. 2018
28. **University of California, Berkeley** - Center for the Built Environment (CBE) Guest Research Lecture and Lunch Talk, Invited Speaker - Berkeley, CA, USA - *Impacting People's Lives Through Data-Driven Innovation in the Built Environment* - Mar. 2018
29. **Delft University of Technology (TU Delft)** - Guest Research Lecture, Invited Speaker - Delft, Netherlands - *Fast, Exploratory, and Scalable: Data Science for Sustainable, High-Performance Buildings* - Nov. 2017

Media

1. **REHVA Journal Article** - *AI-driven Prediction versus Human-driven Judgment in the Built Environment* - Jun. 2025
2. **edX Press Release** - *Building the Future: How This edX Course Is Helping Architects and Engineers Harness the Power of Data* - Nov. 2022
3. **International Building Performance Simulation Association (IBPSA) World Newsletter** - *Data-driven simulation: What's the drive?* - Eleonora Brembilla interviews Laure Itard, Clayton Miller, and Ruchi Choudhary - Oct. 2022
4. **Eco-Business** - *Nurturing greener tenants for more sustainable buildings* - Mar. 2022

5. **ASHRAE Journal Newsletter** - *Searching for the Best Data-Driven Energy Prediction Techniques* - Apr. 2021
6. **Tech in Asia, NUS News, Building Review Journal** - *NUS and Johnson Controls embark on S\$5 million research to co-create people-centric smart building systems* - Mar. 2021
7. **NUS News** - *A Class Of Their Own: Building the future with data* - Mar. 2021
8. **ArchDaily** - *Clayton Miller on Data Science in Architecture: the Academic and Industrial Field are Just Starting* - Oct. 2020
9. **NUS News** - *Learning goes global: NUS launches online courses for learners worldwide* - Oct. 2020

Research and Professional Digital Platforms

- **ORCID Profile, Scopus (Elsevier Profile), Loop (Frontiers Profile), ACM Digital Library Profile, Sciprofiles (MDPI Profile), edX Profile, BUDS Lab Website, LinkedIn, GitHub - BUDS Lab, GitHub - Individual, Stackoverflow, Kaggle, ResearchGate - BUDS Lab, ResearchGate - Individual**

Teaching

Singapore Management University (SMU)

- Course Coordinator, Lecturer, **INTS701: Studying Cities: Research Methods**, Jan. 2026
- Lecturer, **COR3001: Big Questions**, Jan. 2026

National University of Singapore (NUS)

- Course Coordinator, Lecturer (100% of the lectures), and Course Founder, **BPS5229 Data Science for the Built Environment**, 2019 - 2025
- Course Coordinator, Co-Lecturer (70% of the lectures), **BPS5223 Passive Design**, with Assoc. Prof. David Cheong, 2017 - 2024
- Co-Lecturer (23% of the lectures), **BPS5222 Indoor Environmental Quality**, with Prof. Chandra Sekhar, Assoc. Prof. Kwok Wai Tham and Siew Eang Lee (with Assoc. Prof. Eddie Lau 2021-2023), 2017 - 2025
- Co-Lecturer (46% of the lectures), **PF3211 AI Applications for the Built Environment**, with Asst. Prof. Maomao Hu, (with Asst. Prof. Vincent Gan 2022-2023 and Asst. Prof. Yan Ke, 2021 - 2022), 2022 - 2024
- Lecturer and Course Founder, **edX Data Science for Construction, Architecture and Engineering**, Massive Open Online Course (MOOC) 2020-2024 - **43,500+ participants worldwide, 1,830+ Verified Certificates**
- Course Coordinator, Course Co-Founder, Co-Lecturer (54% of the lectures), **IPM/PF1103 Digital Construction**, with Asst. Prof. Vincent Gan (with Asst. Prof. Wang Qian 2017-2022), 2017 - 2023
- Co-Lecturer (46% of the lectures), **PF4501 Total Building Performance**, with Assoc. Prof. Kwok Wai Tham, 2017 - 2020
- Co-Lecturer (23% of the lectures), **BPS5228 Building Materials and Structures**, with Asst. Prof. Kwok Wei Shah, 2017 - 2019
- Co-Lecturer (46% of the lectures), **PF2501/PF2102 Structural Systems**, with Assoc. Prof. Yang Miang Goh, 2017 - 2019

Executive Courses and Workshops

1. **iBuildSG LEAD Milestone Programme: SMU – BCA Advanced Management Programme** - Singapore - *Technology and AI for the Built Environment* - Lecture (Paid Consulting) - Mar. 2026
2. **NUS SCALE Executive Course on Placemaking: Creating liveable and loveable spaces** - Singapore - *Data Science, IoT, Wearables, and other Technologies for Placemaking* - Lecture (Paid Consulting) - Nov. 2020, Jan. 2021, Jul. 2021, Jan. 2022, Apr. 2022
3. **Narxoz University, Narxoz Business School** - Almaty, Kazakhstan - *Big Data for Managers and Decision-makers* - Two Day Workshop (Paid Consulting) - Sep. 2018 and May 2019

4. **Verny Capital** - Astana, Kazakhstan - *Big Data for Managers and Decision-makers - Two Day Workshop* (Paid Consulting) - Sep. 2018
5. **SimAUD 2018 Conference** - Delft, Netherlands - *Python for Building Performance Analysis* - Half-day Workshop - Jun. 2018

Guest Teaching Lectures

1. **Singapore Management University (SMU)**, Guest Lecture - Singapore - *AI and Data Disrupting in the Built Environment? Case Study Discussion of Urban Analytics* - Apr. 2025
2. **Hari Shankar Singhanian School of Business (HSB)**, Guest Lecture - Online - *Urban Analytics Crash Course for DBAs* - Feb. 2025
3. **Technion - Cornell Tech.**, Guest Lecture - Online - *The Human Dimension - Using field-based intensive longitudinal data to characterize humans and buildings* - Jun. 2023
4. **IIT Bombay**, Guest Lecture - Online - *The Role of Open Data and Benchmarking Machine Learning for Built Environment Applications* - Mar. 2023
5. **Aarhus University**, Guest Lecture - Aarhus, Denmark - *The Role of Open Data and Benchmarking Machine Learning for Built Environment Applications* - Oct. 2022
6. **Prince Mohammad Bin Fahd University**, Guest Lecture - Online - *Placemaking using IoT, Wearables and Data Science* - Oct. 2021
7. **National Autonomous University of Mexico (UNAM)**, Guest Lecture - Online and Recorded for the Digital Course *Building with Energy Efficiency and Thermal Comfort in Latin America - Why Building Industry Professionals Should Learn to Code!* - Oct. 2021
8. **University of Texas at Austin**, Guest Lecture - Online - *What do occupants want? Let's ask them using smartwatches and Cozie* - Dec. 2020
9. **Princeton University**, Guest Lecture - Online - *What do occupants want? Let's ask them using smartwatches and Cozie* - Apr. 2020
10. **University of New South Wales (UNSW)**, Digital Cities Guest Lecture - Sydney, Australia - *Introduction to Data Science for Environmental Sensing* - Feb. 2019
11. **Singapore University of Technology and Design (SUTD)**, Guest Lecture - Singapore - *A Feedback Loop to Design* - Nov. 2018

Theses and Dissertations

Ph.D./Master's Degree (Research) Supervisor

1. Cherry Kim – Started primary supervision of **NUS Ph.D.** in Jul. 2025, Transitioned official NUS supervision role to Adrian Chong (but maintained collaborative role in my SMU position) in Aug. 2025
2. Wenhao Zhang – Started primary supervision of **NUS Ph.D.** in Jan. 2024, Passed Qualifying Exam in Aug. 2025, Transitioned official NUS supervision role to Adrian Chong (but maintained primary advisory responsibilities in my SMU position) in Aug. 2025
3. Pradeep Alva – Secondary **NUS Ph.D.** co-supervision started with Rudi Stouffs as primary supervisor in Mar. 2023, Defended successfully in Jun. 2025
4. Yun Xuan Chua - Started primary supervision of **NUS Master's Degree (Research)** in Jan. 2023, Submitted in May 2024, Completed in Nov. 2024
5. Chun Fu – Started primary supervision of **NUS Ph.D.** in Jan. 2020, Passed Qualifying Exam in Sep. 2022, Defended successfully in Mar. 2024, Completed in Jul. 2024
6. Matias Quintana - Started primary supervision of **NUS Ph.D.** in Aug. 2018, Passed Qualifying Exam in Jun. 2020, Defended successfully in Oct. 2022, Completed in Feb. 2023
7. Mahmoud Abdelrahman - Started primary supervision of **NUS Ph.D.** in Aug. 2018, Passed Qualifying Exam in Oct. 2020, Defended successfully in Mar. 2022, Completed in Apr. 2022
8. Irvin Tang Kaijun - Started NUS Ph.D. in Jan. 2017, Transitioned to my primary supervision in Oct. 2017, Transitioned to **NUS Master's Degree (Research)** in Jun. 2021, Submitted MSc. in Jan. 2022, Completed MSc. in Jun. 2022

Internal Ph.D./Master's Degree Qualifying Exam and Dissertation/Thesis Examiner

1. Jean You - **NUS Ph.D.** Qualifying Examiner - May 2025
2. Ivan Poon Kin Ho - **NUS Ph.D.** Dissertation Examiner - Apr. 2025
3. Siyu Cheng - **NUS Ph.D.** Qualifying Examiner - Jan. 2025
4. Eikichi Ono - **NUS Ph.D.** Qualifying Examiner - Jan. 2022, Dissertation Examiner - Jan. 2025
5. Yaonan Gu - **NUS Ph.D.** Dissertation Examiner - Nov. 2024
6. Kai Chen - **NUS Ph.D.** Qualifying Examiner - Oct. 2024
7. Tao Wang - **NUS Ph.D.** Qualifying Examiner - Aug. 2023, Dissertation Examiner - Aug. 2024
8. Yuxin Feng - **NUS Ph.D.** Qualifying Examiner - Aug. 2024
9. Hanjing Zhu - **NUS Ph.D.** Qualifying Examiner - Oct. 2022, Dissertation Examiner - Jul. 2024
10. Feiyang Wei - **NUS Ph.D.** Qualifying Examiner - Jul. 2024
11. Hanzhang Zhan - **NUS Ph.D.** Qualifying Examiner - Dec. 2023
12. Abraham Wu - **NUS Ph.D.** Qualifying Examiner - Dec. 2023
13. Zhuyin Lu - **NUS Ph.D.** Qualifying Examiner - Nov. 2023
14. Xiayi Chen - **NUS MSc.** Thesis Examiner - Oct. 2023
15. Winston Yap - **NUS Ph.D.** Qualifying Examiner - Jul. 2023
16. Pradeep Attavar Alva - **NUS Ph.D.** Qualifying Examiner - Apr. 2023
17. Prashant Anand - **NUS Ph.D.** Dissertation Examiner - May 2019
18. Yiming Liu - **NUS Ph.D.** Qualifying Examiner - Mar. 2019
19. Kuniaki Mihara - **NUS Ph.D.** Qualifying Examiner - Nov. 2018
20. Alexandros Pantazaras - **NUS Ph.D.** Dissertation Examiner - Jun. 2018

External Ph.D./Master's Degree Examiner/Reviewer and/or Committee Member/Co-Advisor

1. Xinyue Wang - **Chalmers University** - Gothenburg, Sweden (Externally Funded for In-Person Attendance) - Ph.D. Dissertation Examiner (Opponent) - Jan. 2026
2. Juliana Croffi Calixto - **University of Adelaide** - Adelaide, Australia (Written Evaluation Only) - Ph.D. Dissertation Examiner - Dec. 2025
3. Roberto Chiosa - **Polytechnic University of Turin (Politecnico di Torino)** - Turin, Italy (Online Oral Defense Attendance) - Ph.D. Dissertation Examiner - Sep. 2025
4. Vineet Kumar Arya - **Massey University** - Auckland, New Zealand (Written Evaluation Only) - Ph.D. Pre-Examiner - Jun. 2025
5. Quinten Carton - **KU Leuven** - Ghent, Belgium - Ph.D. Dissertation Examiner - Internal Examination (Online Oral Defense Attendance) - Feb. 2025, Public Examination (Externally Funded for In-Person Attendance) - Mar. 2025
6. Christian Hepf - **Technical University Munich (TUM)** - Munich, Germany (Online Oral Defense Attendance) - Ph.D. Dissertation Examiner - Feb. 2025
7. Antonio Liguori - **RWTH Aachen University** - Aachen, Germany (Online Oral Defense Attendance) - Ph.D. Dissertation Examiner - Jan. 2025
8. Ben Gottkehaskamp - **Technical University Munich (TUM)** - Munich, Germany (Online Oral Defense Attendance) - MSc. Co-supervisor and Examiner - Dec. 2024
9. Kingsley Nweye - **University of Texas at Austin** - Austin, TX, USA (Online Oral Defense Attendance) - Ph.D. Comprehensive Examiner - Sep. 2024, Ph.D. Dissertation Examiner (Online Oral Defense Attendance) - Nov. 2024
10. Tobias Brudermueller - **Swiss Federal Institute of Technology - Zurich (ETH Zürich)** - Zurich, Switzerland (Externally Funded for In-Person Attendance) - Ph.D. Dissertation Examiner - Oct. 2024
11. Arlinda Bresa - **University of Zagreb** - Zagreb, Croatia (Externally Funded for In-Person Attendance) - Ph.D. Committee Member - Oct. 2021, Ph.D. Dissertation Examiner - Sep. 2024
12. Archana A. P. - **Indian Institute of Technology Madras (IIT Madras)** - Chennai, India (Online Oral Defense Attendance) - Ph.D. Dissertation Examiner - Jul. 2024

13. Azin Velashjerdi Farahani - **Aalto University** - Espoo, Finland (Written Evaluation Only) - Ph.D. Pre-Examiner - Jul. 2024
14. Vu Nguyen - **Swiss Federal Institute of Technology - Zurich (ETH Zürich)** - Zurich, Switzerland (Online Oral Defense Attendance) - Master's Degree Co-Advisor and Examiner - May 2024
15. Jeremy Wong - **University of British Columbia** - Vancouver, Canada (Online Oral Defense Attendance) - Master's Degree Examiner - Dec. 2023
16. Seoyeon Yun - **Swiss Federal Institute of Technology - Lausanne (EPFL)** - Lausanne, Switzerland (Externally Funded for In-Person Attendance) - Ph.D. Dissertation Examiner - Sep. 2023
17. Emmanouil Thrampoulidis - **Swiss Federal Institute of Technology - Zurich (ETH Zürich)** - Zurich, Switzerland (Online Oral Defense Attendance) - Ph.D. Dissertation Examiner - Jul. 2023
18. Máira Afonso de André - **Federal University of Santa Catarina** - Florianopolis, Brazil (Online Oral Defense Attendance) - Ph.D. Qualifying Examiner - Jul. 2021, Ph.D. Dissertation Examiner - Jun. 2023
19. Julien LePrince - **Eindhoven University of Technology (TU Eindhoven)** - Eindhoven, Netherlands (Online Oral Defense Attendance) - Ph.D. Dissertation Examiner - May 2023
20. Kyungmin Lee - **University of Delaware** - Newark, DE, USA (Online Oral Defense Attendance) - Ph.D. Qualifying Examiner - Feb. 2022
21. Benedetto Grillone - **BarcelonaTech (UPC) (Universitat Politècnica de Catalunya)** - Barcelona, Spain (Online Oral Defense Attendance) - Ph.D. Dissertation Examiner - Dec. 2021
22. Till Stoeckman - **Karlsruhe Institute of Technology (KIT)** - Karlsruhe, Germany (Online Oral Defense Attendance) - Master's Degree Co-Advisor and Examiner - Oct. 2021
23. Darryl Fitzgerald - **Murdoch University** - Murdoch, WA, Australia (Written Evaluation Only - No Oral Defense) - Master's Degree Reviewer - May 2021
24. Jonathan Roth - **Stanford University** - Palo Alto, CA, USA (Online Oral Defense Attendance) - Ph.D. Dissertation Examiner - Aug. 2020
25. Marco Savino Piscitelli - **Polytechnic University of Turin (Politecnico di Torino)** - Turin, Italy (Online Oral Defense Attendance) - Ph.D. Dissertation Examiner - Jun. 2020
26. Mauricio Loyola - **Princeton University** - Princeton, NJ, USA (Online Oral Defense Attendance) - Ph.D. Dissertation Examiner - May 2020
27. Eric Teitelbaum - **Princeton University** - Princeton, NJ, USA (Online Oral Defense Attendance) - Ph.D. Dissertation Examiner - May 2020
28. Tao Wang - **University College London** - London, UK (Written Evaluation Only - No Oral Defense) - Master's Degree Co-Supervisor - Jan. 2020
29. Zakia Afroz - **Murdoch University** - Murdoch, WA, Australia (Written Evaluation Only - No Oral Defense) - Ph.D. Dissertation Reviewer - Nov. 2019
30. Cheng Kai Wang - **Delft University of Technology (TU Delft)** - Delft, Netherlands (Online Oral Defense Attendance) - Master's Degree Co-Advisor and Examiner - Oct. 2018

Post-Doctoral Researcher Supervisor

1. Dr. Mario Frei - **Dr. sc. ETH Zürich** - Jan. 2021 - Present (Co-supervised with Bon-Gang Hwang and Daniel Wong from NUS in 2021 and with Filip Biljecki in 2022-2024. Now based in the BEARS organization and co-supervised by Stefano Schiavon in 2024 - Present)
2. Dr. Martin Mosteiro Romero - **Dr. sc. ETH Zürich** - Mar. 2021 - Oct. 2023 (Co-supervised with Rudi Stouffs from NUS)
3. Dr. Matias Quintana - **Ph.D. from NUS** - Oct. 2022 - Mar. 2023
4. Dr. Vasantha Ramani - **Ph.D. from NUS** - Jul. 2021 - Dec. 2023 (Based within BEARS SBB2 Lab - Co-supervised with Kameshwar Poolla from UC Berkeley)
5. Dr. Chuan Fu Tan - **Ph.D. from NUS** - Sep. 2021 - Sep. 2022 (Based within the Johnson Controls Open Blue Innovation Center)
6. Dr. Miguel Martin - **Ph.D. from NUS** - Nov. 2020 - Jun. 2022 (Based within BEARS SBB2 Lab - Co-supervised with Kameshwar Poolla from UC Berkeley)
7. Dr. Prageeth Jayathissa - **Dr. sc. ETH Zürich** - Nov. 2018 - Jan. 2020
8. Dr. Pandarasamy Arjunan - **Ph.D. from IIIT Dehli** - Aug. 2018 - Feb. 2020 (Based within BEARS SBB2 Lab - Co-supervised with Kameshwar Poolla from UC Berkeley)
9. Dr. Jayashree Chadalaawada - **Ph.D. from NUS** - Jan. 2018 - Jun. 2019

Researcher Supervisor

1. Yun Xuan Chua - **NUS Research Assistant** - Feb. 2022 - Aug. 2025 (Co-supervised with Daniel Wong from NUS in 2022)
2. Subin Lin - **BEARS SBB2 Research Engineer** - Sep. - Dec. 2022 (Co-supervised with Kameshwar Poola from UC Berkeley)
3. Mahmoud Abdelrahman - **Research Associate and NUS Ph.D. Student** - Jan. 2021 - Mar. 2022
4. Jonathan Roth - **Research Associate and Stanford University Ph.D. Student** - Aug. 2019 - Jun. 2020
5. Tapeesh Sood - **Research Associate** - Sep. 2017 - Jun. 2019
6. Dongkyu Lee - **Research Associate** - Dec. 2017 - Jun. 2018

Visiting Researcher/Fulbright Student Fellow Hosting

1. Xue Cui - Non-graduating NUS Research Exchange and **Hong Kong Polytechnic University Ph.D. Student** - Feb. - Aug. 2025
2. Desai Wang - **Fulbright Scholar to Singapore** - Sep. 2024 - Jun. 2025
3. Synne Krekling Lien - Non-graduating NUS Research Exchange and **Norwegian University of Science and Technology (NTNU) Ph.D. Student** - Sep. 2024 - May. 2025
4. Ben Gottkehas Kamp - Visiting Researcher and **Technical University Munich (TUM) MSc. Student** - Aug. 2023 - Apr. 2024
5. Sailin Zhong - Visiting Researcher and **University of Fribourg Ph.D. Student and MIT Media Lab Researcher** - May - Jul. 2023
6. Kristi Maisha - **Fulbright Scholar to Singapore and Stanford University Ph.D. Student** - Aug. 2022 - Jun. 2023
7. Antonio Liguori - Non-graduating NUS Research Exchange and **RWTH Aachen University Ph.D. Student** - Nov. 2022 - Apr. 2023
8. Dr. Hussain Kazmi - Visiting Research Fellow and **KU Leuven Post-Doctoral Scholar** - Jan. - Mar. 2023
9. Xiaoyu Jin - Non-graduating NUS Research Exchange and **Hong Kong Polytechnic University Ph.D. Student** - Aug. - Dec. 2022
10. Pierre Garreau - Research Intern and **Ecole des Mines de Saint-Étienne MSc. Student** - Mar. - Aug. 2022
11. Filippo Vittori - Non-graduating Research Exchange and **University of Perugia Ph.D. Student** - Mar. - May 2022
12. Julien Leprince - Non-graduating Research Exchange and **Eindhoven University of Technology (TU/e) Ph.D. Student** - Oct. 2021 - Apr. 2022
13. Ananya Joshi - **Fulbright Scholar to Singapore and Carnegie Mellon University Ph.D. Student** - Aug. 2019 - Jun. 2020
14. Anjukan Kathirgamanathan - **Universitas 21 Visiting Ph.D. Student Exchange Student** - Apr. - May 2019

External Service - Professional

University Service

- **Member of the Faculty Promotion and Tenure Committee (FPTC)**, NUS College of Design and Engineering, Jun. 2024 - Jul. 2025
- **Deputy Head of Communication and Engagement**, Department Management Committee (DMC), NUS Department of the Built Environment (DBE), Jul. 2022 - Feb. 2025
- Member of the **NUS Massive Open Online Course (MOOC) Guidance and Development Learning Community** supported by a Teaching Enhancement Grants (TEG), 2020 - 2022
- NUS-wide **Technology-enhanced Learning (TEL) Showcase Symposium** Presenter - Reaching thousands online! Creating an edX massive open online course (MOOC) - Nov. 2020 (Online)

- Volunteer for Advising Educators in NUS in the **Development of edX MOOC Online Courses** - 2020 - 2023
- Leader of the **SDE4 Learning Trail App** Development for the Opening Ceremony attended by Guest-of-Honor DPM Heng Swee Keat, 2019
- **NUS SDE Workshop for Data Analytics Planner** - *A Post-Spreadsheet World: Tools Easing the Learning Curve for Computational Thinking and Data Analytics* - Sep. 2017
- **International Liaison**, NUS Dept. of Building, 2018

Scientific Community Service and Leadership

- Advisory Board - **EU-funded MuSIC Doctoral Network Project - Multi-sensory solutions for increasing human-building resilience in the face of climate change**, 2022 - 2026
- Non-quorum Voting Member, **ASHRAE Technical Committee (TC) 4.7 - Energy Calculations**, 2021 - 2025
- Observer (Advisory Role) - **EU-funded ADRENALIN Project - dAta-DRivEN smArt buiLdINgs: Data sandbox, competition, implementation**, 2022 - 2025
- **IEA EBC - Annex 79 - Occupant-Centric Building Design and Operation - Subtask 4 Co-Leader** with Zoltan Nagy and Burak Gunay, 2019 - 2024
- **IEA EBC - Annex 81 - Data-Driven Smart Buildings - Activity D3 Leader** - 2020 - 2024
- Resource Person, **BCA Tripartite Facilities Management Implementation Committee (FMIC) - Integrated Smart FM Taskforce**, 2021 - 2023
- Member of the Performance Advisory Committee, **International Well Buildings Institute (IWBI)**, 2021 - 2022

Editorial Boards and Positions

- Editorial Board Member, **Building Simulation** Journal, 2025 - Present
- Editorial Board Member, **Urban Transitions** Journal, 2025 - Present
- Editorial Board Member, **Scientific Data** Journal, 2024 - Present
- Early Career Editorial Board (ECEB) Member, **Building and Environment** Journal, 2022 - Present
- Review Editor, **Frontiers in Energy Efficiency** Journal, 2025 - Present
- Associate Editor, **Science and Technology for the Built Environment** Journal, 2023 - 2025
- Guest Editor, **Energy and Buildings** Journal, Digital twin-enabled building operations and control, 2024
- Guest Editor, **Building Simulation** Journal, Data-driven Building Energy Management Technologies that Integrate Physical Knowledge, 2024
- Guest Editor, **Data-Centric Engineering** Journal, Selected papers from the ACM BuildSys 2023 Conference, 2023
- Guest Editor, **Frontiers in Built Environment** Journal, Innovative Human-Centric Investigations and Technologies for Human Wellbeing and Health in the Built Environment, 2020 - 2021
- Guest Editor, **Energies** Journal, Data Modeling, and Analytics Applied to Buildings, 2020 - 2021

Conference Organizational Leadership and Technical/Scientific Committees

- **Association of Computing Machines (ACM) BuildSys - International Conference on Systems for Energy-Efficient Built Environments**
 - ACM BuildSys 2026, **General Chair**, Inaugural ACM Sustainability Week Co-Chair, Banff, Canada, Jun. 2026
 - ACM BuildSys 2024, **Student Travel Grants Chair** and **Technical Program Committee Member**, Hangzhou, China, Nov. 2024
 - ACM BuildSys 2023, **Technical Program Committee Co-Chair**, Istanbul, Turkey, Nov. 2023
 - ACM BuildSys 2022, **Workshops Co-Chair** and **Technical Program Committee Member**, Boston, USA, Nov. 2022

- ACM BuildSys 2021, **Publicity Co-Chair and Technical Program Committee Member**, Coimbra, Portugal, Nov. 2021
- ACM BuildSys 2020, **Technical Program Committee Member**, Yokohama, Japan (Online), Nov. 2020
- ACM BuildSys 2019, **Publicity Co-Chair and Technical Program Committee Member**, New York, NY, USA, Nov. 2019
- ACM BuildSys 2018, **Poster and Demo Chair and Technical Program Committee Member**, Shenzhen, China, Nov. 2018
- **International Building Performance Simulation Association (IBPSA) Building Simulation Conferences**
 - **Local Organizing Committee Member and Technical Committee Co-Chair, IBPSA ASIM 2026 Conference**, Singapore, Nov. 2026 (Currently in Planning Phase)
 - Building Simulation 2021 (BS2021) International IBPSA Conference, **Scientific Committee Member**, Bruges, Belgium, Sep. 2021
 - eSIM 2020/21 IBPSA-Canada Regional Conference, **Scientific Committee Member**, Online, Jun. 2021
 - Building Simulation 2019 (BS2019) International IBPSA Conference, **Scientific Committee Member**, Rome, Italy, Sep. 2019
 - AIRAH and IBPSA's Australasian Building Simulation 2017 Conference, **Scientific Committee Member**, Melbourne, November 15-16, 2017
- **Local Organizing Committee Member, Indoor Air Conference**, Singapore, Jun. 2026
- **Track Chair for Resilient Smart Cities – Adapting to Climate Change and Transforming the Urban Application Domain**, 11th IEEE International Smart Cities Conference, Patras, Greece, Oct. 2025
- **ACM e-Energy - International Conference on Future Energy Systems**
 - ACM e-Energy 2025, **Technical Program Committee Member**, Rotterdam, NL, Jun. 2025
 - ACM e-Energy 2024, **Workshops Co-Chair and Technical Program Committee Member**, Singapore, Jun. 2024
 - ACM e-Energy 2020, **Technical Program Committee Member**, Melbourne, Australia (Online), Jun. 2020
- **AMLIES - An ACM e-Energy International Workshop on Applied Machine Learning for Intelligent Energy Systems and SAFE-Energy e-Energy Workshop**
 - ACM SAFE-Energy e-Energy Workshop 2024, **Workshop Co-Chair and Technical Program Committee Member**, Singapore, Jun. 2024
 - ACM AMLIES 2021, **Technical Program Committee Member**, Torino, Italy, Jun. 2021
 - ACM AMLIES 2020, **Technical Program Committee Member**, Melbourne, Australia (Online), Jun. 2020
 - ACM AMLIES 2019, **Technical Program Committee Member**, Phoenix, AZ, USA, Jun. 2019
- **ACM DATA - BuildSys/Sensys Workshop on Data: Acquisition to Analysis Conference and Workshops**
 - ACM DATA 2022, **Technical Program Committee Member**, Boston, MA, USA, Nov. 2022
 - ACM DATA 2021, **Technical Program Committee Member**, Coimbra, Portugal (Hybrid), Nov. 2021
 - ACM DATA 2020, **Technical Program Committee Member**, Yokohama, Japan (Online), Nov. 2020
 - ACM DATA 2019, **Technical Program Committee Member**, New York, NY, USA, Nov. 2019
- **Lead Organizer, 9th Expert Meeting of the Annex 79 and OB-22 Symposium**, Singapore, Sep. 2022
- **Scientific Committee Member, 18th Healthy Buildings Europe 2023 Conference**, Aachen, Germany, Jun. 2023

- **ACM BALANCES - BuildSys International Workshop on Big Data and Machine Learning for Smart Buildings and Cities**
 - ACM BALANCES 2025, **Technical Program Committee Member**, Golden, CO, USA, Nov. 2025
 - ACM BALANCES 2022, **Technical Program Committee Member**, Boston, MA, USA, Nov. 2022
 - ACM BALANCES 2021, **Technical Program Committee Member**, Coimbra, Portugal (Hybrid), Nov. 2021
- **Reviewer, ACADIA 2022, Association for Computer Aided Design in Architecture Conference**, Hybrids & Haecceities, Philadelphia, USA, Oct. 2022
- **Technical Program Committee Member, ISC2 2022, 8th IEEE International Smart Cities Conference**, Paphos, Cyprus, Sep. 2022
- **Symposium on Simulation for Architecture and Urban Design (SimAUD) Conference**
 - SimAUD 2021, **Scientific Committee Member**, Online, Apr. 2021
 - SimAUD 2020, **Scientific Committee Member**, Online, May 2020
 - SimAUD 2019, **Scientific Committee Member**, Atlanta, GA, USA, Apr. 2019
 - SimAUD 2018, **Scientific Committee Member**, Delft, Netherlands, Jun. 2018
- **Technical Program Committee Member, IEEE ICPADS 2019, International Conference on Parallel and Distributed Systems**, Mobile and Ubiquitous Computing Track, Tianjin, China, Dec. 2019
- **Reviewer, I3CE 2019, International Conference on Computing in Civil Engineering**, Atlanta, USA, Jun. 2019

Journal Peer Reviewer

Publication Peer Reviewer for **Energy and Buildings**, **Building and Environment**, **Applied Energy** (6), **Journal of Building Performance Simulation** (5), **ACM Transactions on Internet-of-Things** (4), **Building Simulation** (4), **Science and Technology for the Built Environment** (4), **Advances in Applied Energy** (3), **Automation in Construction** (3), **Journal of Building Energy** (3), **Energy and Built Environment** (2), **Energy Efficiency** (2), **International Journal of Human-Computer Interaction** (2), **Solar Energy** (2), **PLOS One** (2), **Frontiers in Built Environment** (1), **Applied Sciences** (1), **Architectural Science Review** (1), **Proceedings of the Institution of Civil Engineers** (1), **Advanced Energy Informatics** (1), **Data-Centric Engineering** (1), **Sustainability** (1), **Sustainable Cities and Society** (1), **Developments in the Built Environment** (1), **Energy & Environment** (1), **Energy** (1), **Scientific Data** (1), **Scientific Reports** (1), **Sustainable and Resilient Infrastructure** (1)

Internal/External Grant Reviewer

- Grants & Fellowships Call 2025, **Fund for Scientific Research-FNRS**, Belgium, 2025
- ENAC Fribourg Grants 2024, **Swiss Federal Institute of Technology - Lausanne (EPFL)**, 2024
- Ministry of Education, Tier 1 Grant, **NUS College of Design and Engineering**, 2024
- Leading House Asia Grant, **Swiss Federal Institute of Technology - Zurich (ETH Zurich)**, 2023
- Research Grants Council Funding, **Research Grants Council (RGC) of Hong Kong SAR**, 2023
- New Frontiers in Research Fund, **Natural Sciences and Engineering Research Council of Canada (NSERC)**, 2022
- Discovery Grant Program, **Natural Sciences and Engineering Research Council of Canada (NSERC)**, 2019
- Collaborative Research and Development Grant, **Natural Sciences and Engineering Research Council of Canada (NSERC)**, 2019
- Open Technology Program, **Netherlands Organisation for Scientific Research (NWO)**, 2018
- Funding Program Application, **Ng Teng Fong Charitable Foundation (NTFCF) Research Funding**, 2018

Professional Service and Leadership

- **American Society of Heating, Refrigerating, and Air-Conditioning Engineers (ASHRAE)**
Associate Member, 2007 - 2012, 2017 - Present
ASHRAE NUS Student Chapter Co-Founder, 2010, and Advisor, 2018 - 2025
Student Member, 2004 - 2007, 2012 - 2017
Singapore Chapter Board of Governors Member and Student Activities Chair, 2011 - 2013
- **International Building Performance Simulation Association (IBPSA)**
IBPSA Singapore Chapter, Founding Member (of chapter refounding in 2018) and Electronic Communications Chair 2018 - Present, Member, 2007 - Present
Co-Founder of the first iteration of the Singapore Chapter, 2012 - 2014
- **Fulbright Association (Singapore)**
Webmaster, 2013 - Present
Executive Committee Member, 2012 - 2013
- **ETH Zurich Singapore Alumni Chapter**
Member, 2019 - Present, Chapter Board Member or Treasurer, 2020 - 2022
- **Association for Computing Machinery (ACM)**
Student Member or Member, 2013 - Present (intermittently)
- **U.S. Green Building Council (USGBC) Nebraska Flatwater Chapter**
Board of Directors, 2008 - 2009
- **Society of American Military Engineers (SAME)**
Member, May 2007 - 2010
Student Member, 2005 - 2007
Student Chapter President, 2005 - 2006
- **Architectural Engineering Institute (AEI)**
Student Member, 2005 - 2007

Professional Registrations

- **Singapore Certified Energy Manager (SCEM)** - Jul. 2014
- **Singapore Building and Construction Authority (BCA) Certificate on Measurement and Verification of Chilled Water Plant Efficiency** - Sep. 2011
- **Mechanical Engineer-In-Training (EIT)** in the State of Nebraska, USA (No. E-8582) - May 2007
- **LEED Accredited Professional (LEED AP)** by the U.S. Green Building Council (USGBC) - 2007